

Criterion A

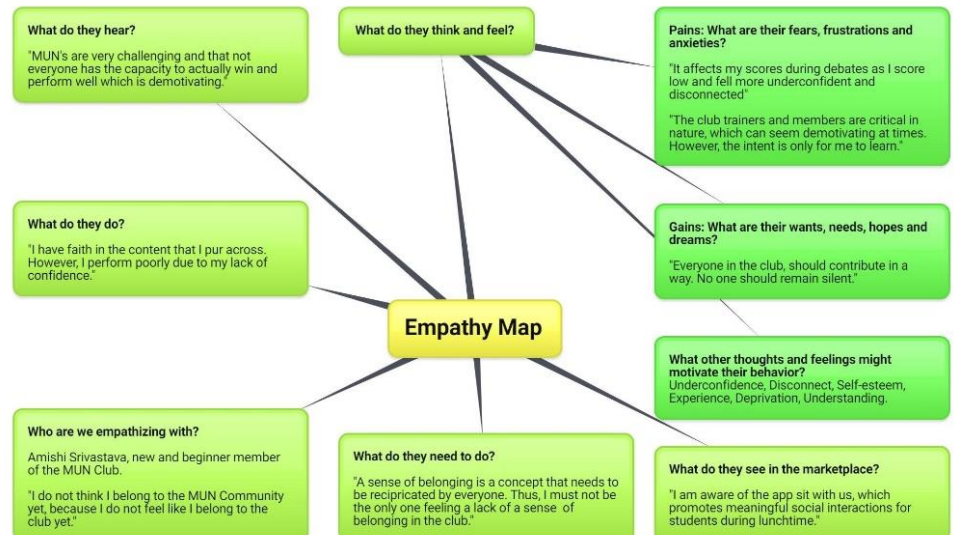
Strand 1: Explain and Justify the need for a solution to a problem for a specified client/target audience.

Client Problem and Cause

My client **Amishi Srivastava** is a **new and beginner member** in the Model UN Club and feels a **lack of a sense of belonging**.

Evidence of problem: Primary Sources

In order to establish the cause of the lack of a sense of belonging, I drew up **an empathy map** (Refer to Figure 1.0) under the guidance of my client that allowed me to understand the **5 W's and 1 H** of her problem.



Evidence of problem: Secondary Sources

Upon further reflection and initial research (Refer to Appendix A.2), I comprehended that a lack of social interaction *deprives her from building connections or bonds with other members of the club* that creates a lack of a sense of belonging. Factors such as low self-confidence and self-esteem fuel her problem.

Figure 1.0

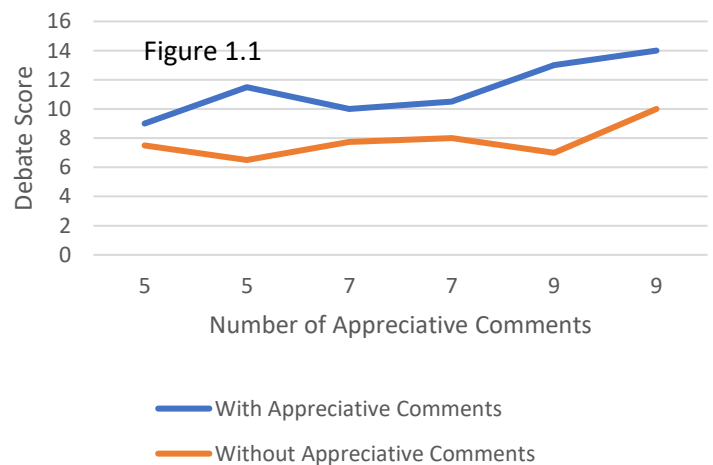
Need for a solution

The Model UN Club debates are evaluated upon the following criteria (Refer to Appendix A.1.2). The lack of a sense of belonging causes my client to score low (Refer to Appendix A.1.1) during debates in Strategy and Style, as components such as Lobbying (more commonly known as diplomatic negotiation) are extremely difficult to demonstrate without a sense of community and belonging.

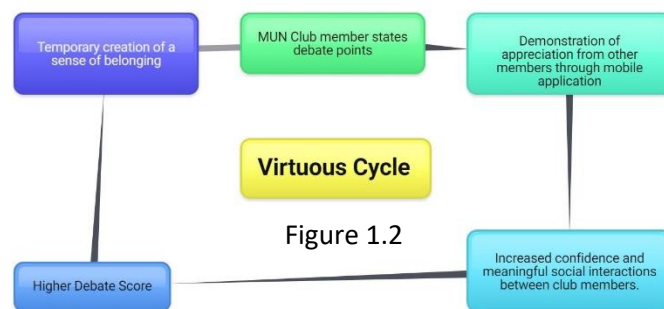
Theory Behind Potential Solution

According to Maslow's hierarchy of needs, belonging comes from a perception of quality, meaning and satisfaction with social connections (**Cherry**). Thus, I theorized that positive reinforcement and appreciation can lead to the creation of a sense of belonging which promotes synergy within the club. In order to prove this theory, I conducted an experiment (Refer to Appendix A.1.2) during an MUN Club meeting where we established the **independent variable** as appreciate comments and the **dependent variable** as the debate score. In addition, I kept a controlled group to achieve fair and accurate results. The results of the experiment demonstrated a positive linear correlation between the number of positive comments and debate score (Refer to Figure 1.1). I then also conducted individual interviews with the club members (Refer to Appendix A.1.3) and gained responses that demonstrated meaningful and satisfactory social interactions during the experiment. Subsequently, demonstrating synergy as a result of a sense of belongingness.

Potential Solution



Therefore, I intend to create a **mobile application** that replicates the ‘two fingers tap of appreciation’ during debates that would cater to Lobbying points and help create a sense of belonging in the MUN club. In addition, during MUN conferences, Social Media application such as WhatsApp and Instagram are typically used to form BLOCS (is a group of delegates who are working together on one or more documents or amendments and also vote together). Thus, making the medium of communication efficient. To predict the effects of the possible solutions, I created a **virtuous cycle (Refer to Figure 1.2)** that demonstrates the perpetual cyclic benefits of creating such a product.



Global Context and Chosen Exploration

The proposed solution connects to the global context of “Identities and Relationships” as it delves into the exploration, “Identity formation, self-esteem, status, roles and role models’ through creating a sense of belonging that would promote synergy in the form of exemplary performance within the MUN Club.

Key Concept and Related Concepts

The proposed solution addresses the key concept as it promotes the social development of club members through creating a sense of belonging. The related concept of collaboration is addressed as a sense of belonging can only be established if reciprocated by each and every member of the club.

Statement of Inquiry

Through creating the proposed solution, it predicted by the virtuous cycle to develop a sense of belonging which will promote synergy within the MUN Club members.

ATL Skills

- Research and Thinking Skills: To brainstorm and develop a product that caters to all the needs of my client
- Communication and Social Skills: To be able to empathize with my client’s problem
- Self-management Skills: To create realistic and feasible goals that I can complete within a given time frame and work independently

Strand ii. Identify and prioritize primary and secondary research needed to develop a solution to the problem

What is the purpose of identifying and prioritizing primary and secondary research?

The purpose of this task is to help in the planning stage, where I would be required to **prioritize elements researched**. For example, for some apps the aesthetics are prioritized over the functionality or visa-versa. In addition, it makes the **researching process easier** by breaking it down into sub-sections. This as a result gives **detailed** and **specific results**. In addition, an **OPVL Analysis (Refer to Appendix A.3)** will be conducted in order to test the credibility and value of the sources.

P: Primary | S: Secondary, Priority: 1 being lowest | 4 being highest

Client Problem				
Question	Type	Source	Justification	Priority and Deadline
What are the causes of a lack of a sense of belonging?	P	Client Interview + MUN Expert Interview + Scientific Study (Refer to Appendix A.2.1)	In order to come up with a feasible solution, it is important to empathize with the client and understand their problem to fully address it. In addition, it is important to gain knowledge from an expert which would provide insight into the subject from someone who has experienced something similar to the client. Lastly, a scientific study	3/4 5.12.20

			would provide credibility to my research. Thus, addressing the need for both primary and secondary sources.	
What are the short-term effects of a lack of a sense of belonging?	S	Client Interview+ MUN Expert Interview + Article (Refer to Appendix A.2.2)	To prevent the negative effects on my client and target audience, it is imperative that I understand what the effects of a lack of a sense of belonging are. A client interview caters to my client's individual needs. The expert interview give insight into an evolved and detailed response from what my client gave. The article from a reliable source quotes anecdotal references which provide a variety of opinions.	1/4 5.12.20
What are the long-term effects of a lack of a sense of belonging?	P + S	Client Interview + MUN Club Statistics + Exit Form + Article + Scientific Study (Refer to Appendix A.2.3)	The MUN Club statistics would provide quantitative data to support the client's statement that numerous members leave the club. The scientific study provides a prediction of what the lack of a sense of belonging can cause in the predicted future. In addition, the exit form submitted by member's who left would provide testimonial for the reason they left. The questions are divided into long- and short-term effects as they differ in scale.	2/4 6.12.20
Did trainers of the club feel a lack of a sense of belonging when they joined the MUN Community? If yes, how did they tackle it?	P	MUN expert Survey (Refer to Appendix A.2.4)	It is important to understand whether my client is the only one affected by such a problem. Therefore, by asking the trainers who were once in the same position would aid to understanding how they tackled the problem.	2/4 6.12.20

Concept

Question	Type	Source	Justification	Priority
Does appreciation foster a sense of community and belonging temporarily or perpetually?	P+S	Scientific Experiment + Article (Refer to Appendix A.2.5)	The scientific experiment provides me with quantitative and qualitative data of my own that would support my hypothesis and help determine the outcome. The article acts as ethos to help support the intended purpose for the proposed solution.	3/4 7.12.20
What role do Maslow's hierarchy of needs play in fostering a sense of belonging?	P+S	Psychology Expert + Textbook (Refer to Appendix A.2.6)	To provide scientific backing to my proposed solution, I provided expert insight that would help break down the content I received from the textbook.	1/4 7.12.20

Functionality

Question	Type	Source	Justification	Priority
What tools can we use to foster appreciation within a club?	S	Scientific Experiment + Article (Refer to Appendix A.2.7)	By understanding what tools foster creative, I can build a foundation for the concept of the proposed solution.	2/4 8.12.20
Are there any existing tools used by the MUN community or similar clubs/organization to foster a sense of community or belonging?	P+S	Client Interview + MUN Expert Interview + Articles (Refer to Appendix A.2.8)	By exploring tools that clubs and organizations such as the MUN Club use, I can identify elements that I would like to incorporate and what to avoid. In addition, it provides insight into existing products in the market.	1/4 8.12.20

Aesthetics

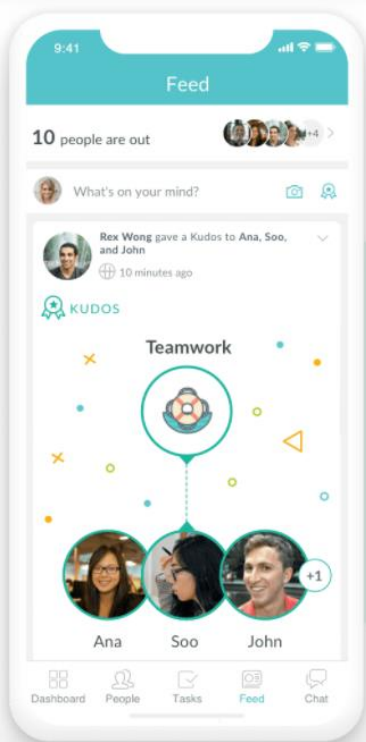
Question	Type	Source	Justification	Priority
Does the application's look and feel affect the intended purpose?	S	Book (Refer to Appendix A.2.9)	This will help me understand how to prioritize functionality and aesthetic. In addition, as my proposed solution's functionality is quite simplistic, I will be able to understand what other factors contribute to its outcome.	1/4 9.12.20

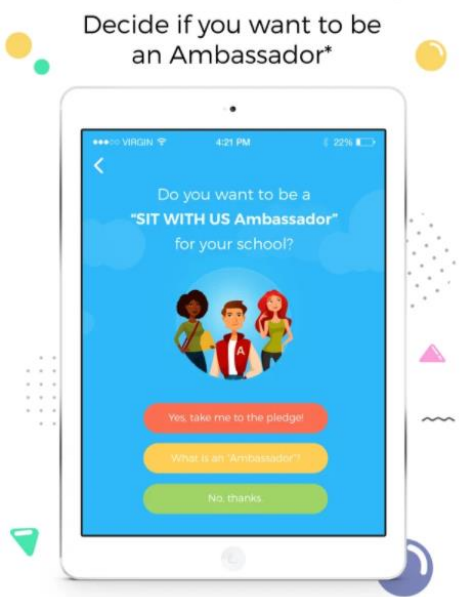
What design trends have surfaced during 2019 and 2020?	S	Articles (Refer to Appendix A.2.10)	This will help me identify what elements to incorporate that have been popular in the past few years.	1/4 9.12.20
What color schemes and fonts promote productivity, gratitude and appreciation?	P+S	Art Expert Interview+ Articles + Survey (Refer to Appendix A.2.11)	This will help me identify the most scientific and user-based appeal to various colors and fonts. It will also me to give more specific options for my client to pick.	2/4 9.12.20

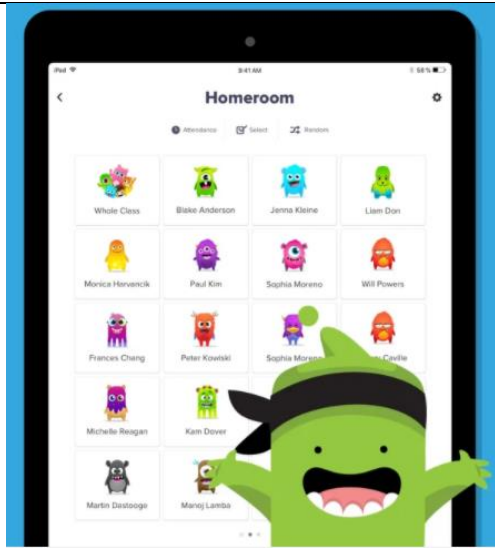
Software				
Question	Type	Source	Justification	Priority
What mobile application development tools exist?	Primary + Secondary	Computer Engineer Interview + Article (Refer to Appendix A.2.12)	In order to decide what development tool will be suitable for my mobile application, I would need to research into all the possible tools that exist. The expert interview would provide me insight into which tool the individual prefers and why.	4/4 10.12.20
What algorithm can be used to provide objective scoring?	Primary + Secondary	MUN Club Expert Interview+ Articles (Refer to Appendix A.2.13)	Another purpose of the proposed solution will be to provide scores that are objective and not biased. By researching into this, I can find the most optimum way to provide lobbying points using the proposed solution.	¾ 10.12.20

Strand iii. Analyze a range of existing products that inspire a solution to the problem

Note: Each aspect of the product is rated out of 10 (1 being lowest, 10 being highest) to produce both quantitative and qualitative results.

Product 1	Parameter	Analysis
 HR Cloud Employee App <u>Why did I choose to analyze this application?</u> The mobile application is used by leading brands such as; The beautiful group, Veolia, Osmose, Spark power corporations etc. Total Score: 44/60	Aesthetics: 7/10	<i>Strengths:</i> The mobile application uses a pastel blue (draw attention, inspire trust, suggest precision, communicate consciousness and stimulate productivity) and white color scheme majorly with smaller accents of pastel green, orange and yellow. The application is minimalistic and uses custom made graphics that are design trends that have surfaced during 2019 and 2020. <i>Weaknesses:</i> The user interphase can appear to be cluttered due to the high number of animations. This can reduce user focus.
	Cost: 5/10	The platform has 4 pricing plans with pricing starting at \$6 per employee per month. <i>Weakness:</i> This may be costly for smaller firms.
	Consumer: 8/10	<i>Strengths:</i> HR Cloud targets companies of all sizes in many industry verticals, such as retail, health care, hospitality, finance and technology. Ex. MPP Infusion and Multispecialty Physician Partners. The pages are customized for better user experience.
	Environment and Ergonomics: 7/10	<i>Weaknesses:</i> Some short comings that the app has are that payroll, attendance management and learning management features are not built into the software.
	Size and Safety: 8/10	<i>Strengths:</i> The application uses the I-9/E-Verify functionality, which allows Human Resources of the organization to verify their employee's eligibility. <i>Weaknesses:</i> This process is inefficient due to its manual nature.
	Functionality: 9/10	<i>Strengths:</i> The mobile app offers peer-to-peer recognition in order to reward their accomplishments that would promote a culture of productivity. It has other features such as a company directory, calendar for employee events and project deadlines.
	Analysis Summary (<i>What would I take inspiration from</i>)	The application is customized as per the firm or organization using it. For example, a hospital or a technology firm. The application uses 'peer-to-peer' recognition which is a form of a reciprocated sense of belonging. Thus, I would take inspiration majorly from its functionality and tailor it to an MUN environment.

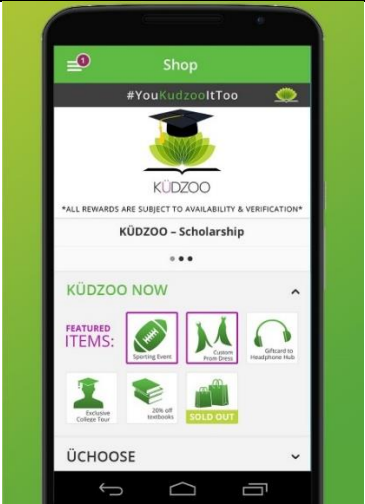
	for the proposed solution?)	
Product 2	Parameter	Analysis
 Sit with Us <u>Why did I choose to analyze this application?</u> My client mentioned the application during her interview and appreciated the intent behind it (Refer to Appendix A.2.8). Total Score: 35/60	Aesthetics: 4/10	<i>Strengths:</i> The product also majorly uses a blue and white palate with red, yellow and green accents. It heavily relies on animations due to its simplicity. <i>Weaknesses:</i> There use of negative space makes the user interphase appear 'outdated' and disregards modern design trends.
	Cost: 8/10	<i>Strengths:</i> The app is completely free. <i>Weaknesses:</i> However, only available on the app store for IOS devices.
	Consumer: 7/10	<i>Strengths:</i> Caters to teachers and students of schools to initiate conversation on the lunch seating ethic. <i>Weaknesses:</i> The user interphase is generic to both teachers and students.
	Environment and Ergonomics: 6/10	<i>Strengths:</i> It provides an authentic way to connect with others, combats isolation, and encourages an inclusive school environment. <i>Weaknesses:</i> There is no way to verify whether the seating arrangement was executed.
	Size and Safety: 3/10	<i>Strengths:</i> The app is 110.5 MB and requires iOS 9.0 or later. In terms of registration, it requires either a Facebook account or an email address and school information. <i>Weaknesses:</i> The chat feature is not moderated. Therefore, there is a risk of cyber-bullying and harassment. In addition, the app doesn't verify user information or age, meaning any and all kinds of people could use it to spot potentially vulnerable children
	Functionality: 7/10	<i>Strengths:</i> The app assigns students' other students to sit with at lunch randomly to create new friends and social circles within their own school. It also allows the student to chat with their prospective lunch 'buddy'. To add a lunch event at a school, a sit with us student ambassador or teacher must add an event on the app calendar. Students will then be assigned lunch 'buddies' for that event.
Analysis Summary (What would I take inspiration from for the proposed solution?)		The app sit with us utilizes a color scheme whose purpose is to draw attention, inspire trust, suggest precision, communicate consciousness and stimulate productivity. These are virtuous effects that would benefit my client and ultimately promote a sense of belonging. Therefore, I will take inspiration from its color scheme.
Product 3	Parameter	Analysis
Class Dojo <u>Why did I choose to analyze this application?</u> Over 95 % of schools in the United States use Class Dojo to reward student behavior and work. In addition, my client said that she used the app in primary school (Refer to Appendix A.2.8).	Aesthetics: 9/10	<i>Strengths:</i> The product also majorly uses a blue and white palate with multicolored alien avatars that are animated. All elements are rounded at the edges to provide a modern and minimalistic look. The application also uses animations to make it more interactive. For example, each student corresponds to an avatar that is animated to do certain actions.
	Cost: 10/10	<i>Strengths:</i> The app is completely free and is available as a website as well. It is available for both IOS and Android devices.
	Consumer: 8/10	<i>Strengths:</i> It is a management platform where teachers can record and track student behavior,



Total Score: 52/60

	facilitate classroom activities, curate student portfolios, and communicate with parents.
Environment and Ergonomics: 8/10	<i>Strengths:</i> Teachers should be sure to emphasize positive reinforcement and use the tool's public features in ways that support students' privacy and dignity. Even in light of teachers' best efforts, some students might still interpret the system in more of an extrinsic way. The program's effectiveness depends largely on responsible and consistent use.
Size and Safety: 8/10	<i>Strengths:</i> The app is 110.5 MB and requires iOS 9.0 or later. ClassDojo's policy states that personal information collected from students is never used for behavior targeted advertising to students by ClassDojo or by any third-party.
Functionality: 9/10	<i>Strengths:</i> The application and website allow teachers to track their student's behavior, facilitate classroom activities, build their portfolios and communicate with their parents. Student sign-in is simple with a QR code, link, or Google account. From the teacher dashboard, teachers can create original activities or use the toolkit to post class announcements, conduct warm-ups, create random groups, and more. Depending on how they choose to set things up, teachers can share information privately or publicly, often displayed to the class on a screen or interactive whiteboard.
Analysis Summary (What would I take inspiration from for the proposed solution?)	The application uses animations to increase interactivity between the user and the graphic interphase. In addition, the minimalistic and modern elements contribute to its aesthetic appeal.

Product 4	Parameter	Analysis
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Kudzoo
 Why did I choose to analyze this application?
 The application was listed under a Forbes list and received approximately 1.3 million dollars in funding in 2015.

Note: There has been an exponential decline in the application's popularity after 2015 and

Aesthetics: 4/10	<i>Strengths:</i> The product also majorly uses a green and white color scheme which intends to restore energy, promote growth, nurture and rejuvenate the user. <i>Weaknesses:</i> However, the screen appears to be extremely claustrophobic due to element placement. It also uses Skeuomorphic design that is outdated.
Cost: 5/10	<i>Strengths:</i> The app is completely free. <i>Weaknesses:</i> However, is only available in the United States as of right now. In addition, they are in a financial deficit as they provide monetary rewards.
Consumer: 3/10	The application's target audience are students in high school. <i>Weaknesses:</i> There are no personalized elements in the UI.
Environment and Ergonomics: 4/10	<i>Weakness:</i> The app works on the business model where they pull rewards from gift card companies and different deal aggregators as well to get them to students. <i>Strength:</i> They are currently in conversation with firms they intend to partner with.
Size and Safety: 4/10	<i>Weakness:</i> It starts off by taking a picture of your report card, uploading it into the app and then it's

therefore, this application is being analyzed for its disadvantages and what to avoid. Total Score: 26/60		reviewed by Kudzoo staff. After that they approve or deny it. If it's approved you get Kudzoo Cash (more or less reward points) that you can redeem in the shop section of the app. This process is inefficient.
	Functionality: 6/10	<i>Strength:</i> Kudzoo provides an app for students above 13 years of age to get rewards for performance in school. Students can submit their report card for Kudzoo Cash that can be used for deals and discounts for giveaways, scholarships and events. Kudzoo Cash can also be earned by attendance, grade improvement and answering daily SAT & trivia questions.
	Analysis Summary (What would I take avoid for the proposed solution?)	Claustrophobic and cluttered element placement should be avoided. In addition, the usage of Skeuomorphic design should be avoided.



Strand iv. Develop a detailed design brief, which summarizes the analysis of relevant research.

Research Planning

I began by empathizing with my client's problem through a survey and an empathy map (**Refer to Criterion A; Strand i and Appendix A.2**) and was successfully able to identify the causes and effects of a lack of a sense of belonging within the MUN Club. I also conducted a scientific experiment that justified the need for a solution to my client's problem. The results demonstrated a positive correlation between appreciative comments and the members debate scores which reflected synergy and a development of a sense of belonging. I then created a research table that would help me identify information I would need to create the proposed solution.

Primary sources

The primary sources of information for research were obtained from a client interview, expert opinions and target audience surveys. The client interview established the cause and effect of her problem, alongside existing products in the market she has observed that aim to create a sense of belonging. I gathered expert opinions from psychology and visual art experts, alongside a computer engineer and experienced MUN participant. The psychology expert's knowledge provided precedent for my client's problem and explained the role of Maslow's hierarchy of needs (**Refer to Appendix A.26**). The visual art expert aided in gathering knowledge about color schemes and how they affect the aesthetics of a product (**Refer to Appendix A.2.11**). The computer engineer provided knowledge on the various types of mobile application development tools that would aid me when developing the application in Criterion C (**Refer to Appendix A.2.12**). The MUN expert validated my client's problem by stating her experience as a beginner MUN participant. She then also stated her observations as the MUN Club trainer (**Refer to Appendix A.2**).

Secondary Sources

The secondary sources of information helped inquire mainly about the functionality of the proposed solution and validating the information received from the primary sources of information. In addition, research into existing products in the market that aim to address a lack of a sense of belonging provided my knowledge on what elements to incorporate and avoid in my product (**Refer to Criterion A; Strand iii.**). By evaluation the advantages and limitations of the products, I identified the following information;

The feature of peer-to-peer appreciation in the mobile application "HR Cloud", would inspire the functionality of the proposed solution. It would also cater to the lobbying score of the user based on an algorithm that calculates that scores the user out of the total number of participants during the debate (**Refer to Appendix A.2.12**). In addition, a mobile application is the most suitable medium as social Media application such as WhatsApp and Instagram are

typically used during MUN conferences. The color scheme would be composed of pasted blues with accents of green, yellow and orange that are similar to the product “Sit with Us”. In addition, due to the simplicity of my app, I would be creating animations similar to “Class Dojo” that make the application GUI more dynamic. The application would also have rounded elements to provide a modern and minimalistic aesthetic which is similar to “Class Dojo” and is a popular design trend in 2019 and 2020 (**Refer to Appendix A.2.10**). The cost of the application will be completely free as there is no need to generate revenue unlike the application Kudzoo.

To determine the credibility and value of the source, I conducted OPVL analyses of the sources (**Refer to Appendix A.3**).

Proposed Solution

The mobile application will have three sections; Sign Up/Log In, Home, Kudos Page and Statistics. Club member’s will be able to sign up using their email ID and a password and will require approval to log in. Upon logging in, the member will be directed to the Home page where they will wait for the trainers to start a debate and can view their own statistics. The ‘Kudos’ page is where they will be able to appreciate the points made by the members debating by tapping on a button. The 2 members debating will be able to view the number of ‘Kudos’ they receive in real time. Upon competition of the debate, the observing members will be directed to the home page where they can see the results of the debate and their own statistics.

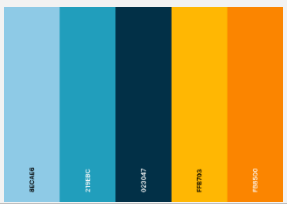
Lastly, I looked at various methods to create the application (**Refer to Appendix A.2.13**); Coding Java with Android Studio, or using online app developing tools such as Build fire, MIT app inventor and Thunkable.

Sustainable Development Goal

Therefore, the product that I intend to create will be able to help address the chosen sustainable development goal of “Good Health and Wellbeing” as it promotes the growth of one’s mental health that thrives of social interactions and positive reinforcement.

Criterion B

Strand i: Develop detailed design specifications, which explain the success criteria for the design of a solution based on the analysis of the research

Aesthetics				
	Design Specification	Justification of Requirements	Testing Method	Constraints / Considerations
A.1	The font ‘Roboto’ must be used in the following sizes;  Headings: 25 - 30 Sub-headings: 20- 25 Paragraphs: 18 - 20	The following font styles were selected originally from popular font styles in 2020 according to my criterion A research (Refer to Appendix A.2.11). They were then narrowed down to these fonts by conducting a survey (Refer to Appendix A.2.11)	A user trial will be conducted and the processes used in the creation will simply be cross-checked.	Thunkable does not have certain fonts available on both IOS and android devices.
A.2	The following color scheme must be used; 	The following color scheme was taken inspiration from the application “Sit with Us” (Refer to Criterion A; Strand iii, Product 2). It was then narrowed down by conducting a survey (Refer to Appendix A.2.11)	(Qualitative data)	
A.3	Elements such as buttons and boxes must be curved at the edges by setting the border radius to either 50 or 75 for a modern and minimalistic aesthetic.	This is a common stylistic feature in “Class Dojo” (Refer to Criterion A; Strand iii, Product 3) and is a popular design trend in 2019 and 2020 (Refer		

		to Appendix A.2.10) according to my research.	
A.4	There must be custom animations that make the application more dynamic.	The simplicity of the app's functionality can be improved upon by adding features that make it more dynamic.	Thunkable has a preset which allows used to import .json files from Lottie Files as free animations.

	Consumer		
	Design Specification	Justification of Requirements	Testing Method
B.1	The application must cater to both the trainers and trainees of the club, by having different permissions.	Both the trainers and trainees cannot have similar permissions and functionality. Therefore, the application must cater to both types of users.	A user trial will be conducted to see whether it caters to both types of users.

	Ergonomics and Environment		
	Design Specification	Justification of Requirements	Testing Method
C.1	The application must increase members scores during debates.	As seen during the experiment conducted (Refer to Appendix A.1.3), there is a correlation between a debate score and a sense of belonging created from appreciation.	Analysis through graphical data collected after 1 week of using proposed solution.
C.2	The application must promote synergy and a sense of belonging.	The purpose of the application is to resolve the lack of a sense of belonging.	User interview to interview whether synergy and a sense of belonging was promoted.

	Safety and Size		
	Design Specification	Justification of Requirements	Testing Method
D.1	User's must sign in using their school email id's	As the club is under the school's supervision, linking the application to the member's school email ids will make it more convenient.	Performance Test
D.2	The application must not share any user data with external parties of 3rd parties.	The MUN Club scores are confidential to the members and trainer, to motivate member and promote a sense of belonging.	A risk assessment of the application will be conducted to test its safety.

	Functionality			
	Design Specification	Justification of Requirements	Testing Method	Constraints / Considerations
E.1	The Application must be divided into the trainers and the trainees' section, with a common sign in page.	Both the trainers and trainees cannot have similar permissions and functionality. Therefore, the application must cater to both types of users.	A user trial will be conducted to see whether it caters to both types of users.	

E.2	The trainers must have the following screens; - Home - Debate settings	As mentioned, while researching (Refer to Criterion A; Strand iv) , the application must have a trainer and trainee section.	Performance testing, which is quantitative in nature will be conducted.	Some splash screen will be added in between to aid the flow of transition between the main pages.
E.3	The trainees must have the following screens; - Home - Kudos			Some splash screen will be added in between to aid the flow of transition between the main pages.
E.4	The application must be available on both IOS and Android devices.			Some splash screen will be added in between to aid the flow of transition between the main pages.

Success Criteria: Each criterion will pass if the expected output is achieved in performance testing

General			
What is being tested	Testing Method	Input	Expected Output
App Start-Up	Robotium	Click on Mobile Application Icon	1-2 Seconds
Battery Time while using an app	Robotium	Using application for 1 round of debates Trainer: 2 minutes for log in + setup and 5 minutes for debate. Trainee: 1 minute for log in and 5 minutes for debate.	10%
Memory Consumption	Robotium		5%
Hardware/Software Variation	Running application various IOS and Android Devices; 1. iPhone X (IOS) a. Resolution: 1792x828 pixel b. A12 Bionic chip, Hexa-core c. RAM: 3 GB 2. Samsung Galaxy S10 (Android) a. Resolution: 2280 x 1080 b. Processor: Octa-Core c. RAM: 6 GB 3. Samsung Galaxy A52 (Android) a. Resolution: 2280 x 1080 b. Processor: Octa-Core c. RAM: 8 GB 4. iPhone 8 (IOS) a. Resolution: 750 x 1334 pixel b. A11 Bionic chip, Hexa-core c. RAM: 2 GB	Using application for 1 round of debates	Functions as mentioned in Design Specification.

Log In Screen			
What is being tested	Testing Method	Input	Expected Output
Log In screen Validity	Inputting various user log in values	Note: Corresponding means that the email address and password are of the same user. Invalid meaning that they do not exist in the Airtable database. 1. Valid and Corresponding Email Address and Password. Email Id: studentx@school.com	1. Navigate to Trainer or Trainee Home Page.

		Password: 1987 2. Invalid Email Address and Password. Email: abcd@school.com Password: 1923 3. Valid Email and Invalid Password. Email: studentx@school.com Password: 1287 4. Invalid Email and Valid Password. Email: abcd@school.com Password: 1987 5. Valid, but not corresponding Email Address and Password. Email: studenty@school.com Password: 1987	2. Error Message 3. Error Message 4. Error Message 5. Error Message
Log In Screen Navigation	Inputting various user log in values	If the user Email ID and Password are valid then; 1. Trainer Email Address and Password Email Id: studentz@school.com Password: 1989 2. Trainee Email Address and Password Email Id: studentx@school.com Password: 1987	1. Navigate to Trainer Home Page 2. Navigate to Trainee Home Page
Response Time of Elements upon starting	Robotium	Click on Mobile Application Icon	0.5 Seconds

Trainer Home Page Screen			
What is being tested	Testing Method	Input	Expected Output
Response Time of Elements upon starting	Robotium	Click on Log In Button on log in screen.	0.5 Seconds
User Image, Name, Grade, Level, Rank	Observation	Click on Log In Button on log in screen.	Shows corresponding user image, name, grade, level and rank with email address from the Airtable.
Log Out button Navigation	Observation	Clicking on Log Out button	Navigate to Log In Screen
Start Debate Navigation	Observation	Clicking on Start Debate button. For scenario 2 (view in expected output), use 2 separate trainer accounts; Email Id: studentz@school.com Password: 1989 Email Id: studenta@school.com Password: 1593	1. If a debate has not started; Navigate to Debate Settings Page 2. If a debate has been started by another trainer; Stay on Home Page

Debate Settings Page Screen			
What is being tested	Testing Method	Input	Expected Output
Response Time of Elements upon starting	Robotium	Click on Mobile Application Icon	0.5 Seconds
Disabled buttons	Observations	Click on Select Debater 2, Start and End Button	None, as buttons should be disabled.

Start Debate Navigation	Observation	Use 2 separate trainer accounts; Email Id: studentz@school.com Password: 1989 (Starts Debate) Email Id: studenta@school.com Password: 1593	If a debate has been started by another trainer; Navigate to Home Page
Select Debater 1 Button	Observation	Click on Select Debater 1 Button	Show List Viewer
Select Debater 1 List Viewer	Observation	Select 'Student X'	Hide List viewer and enable select debater 2 button.
Select Debater 2 Button	Observation	After selecting debater 1, Click on Select Debater 2 Button	Show List Viewer
Select Debater 2 List Viewer	Observation	Select 'Student Y'	Hide List Viewer and enable Start Button.
Hide Debater 1 name from Select Debater 2 List Viewer	Observation	Select 'Student X' as Debater 1 and try the same for Debater 2	Not visible, as when selecting debater 2, the name of the previously selected should not be visible.
Start Button	Observation	After selecting debater 2, Click on Start button.	Disable Start Button, enable end button.
End Button	Observation	After clicking on Start Button, Click on End button.	Navigate to Home Screen
Back Button	Observation	Click on Back Button	Navigate to Home Screen

Trainee Home Page Screen			
What is being tested	Testing Method	Input	Expected Output
Response Time of Elements upon starting	Robotium	Click on Mobile Application Icon	0.5 Seconds
User Image, Name, Grade, Level, Rank and graph	Observation	Click on Log In Button on log in screen.	Shows corresponding user image, name, grade, level, rank and graph with email address from the Airtable.
Log Out button Navigation	Observation	Clicking on Log Out button	Navigate to Log In Screen
Debate Started	Observation	Use 2 accounts to log in; Trainer Email Id: studentz@school.com Password: 1989 (Use to start Debate) Trainee Email Id: studentx@school.com Password: 1987	If debate has begun; Navigate to KUDOS Page

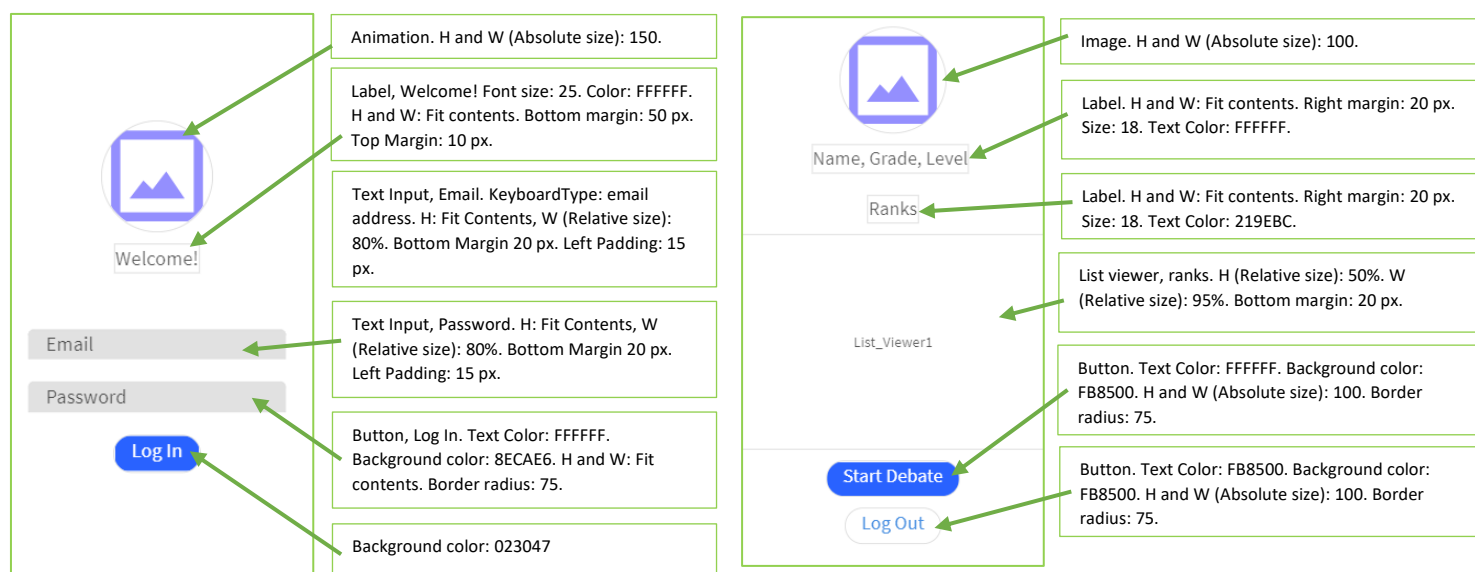
KUDOS Screen			
What is being tested	Testing Method	Input	Expected Output

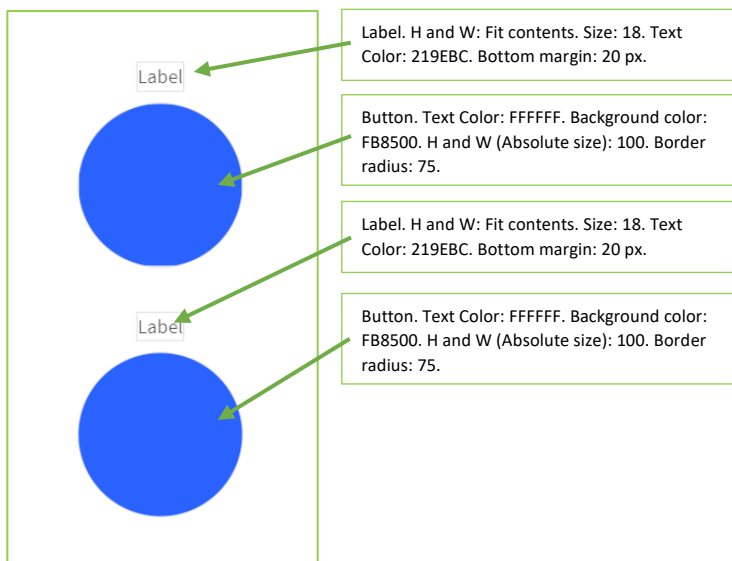
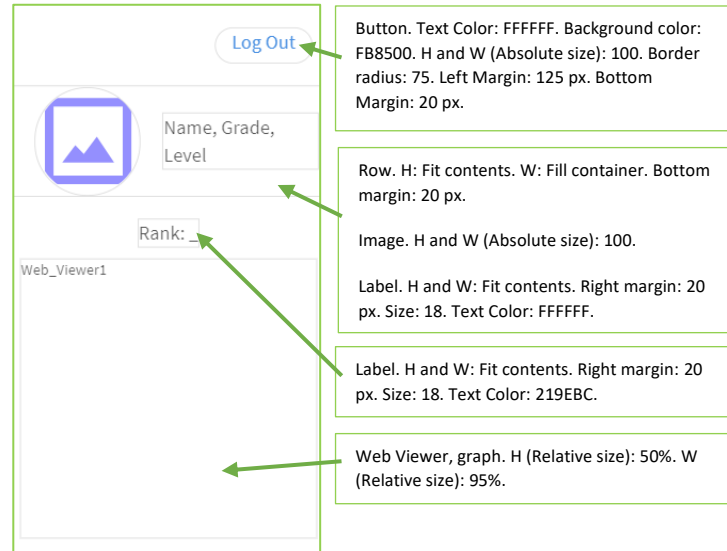
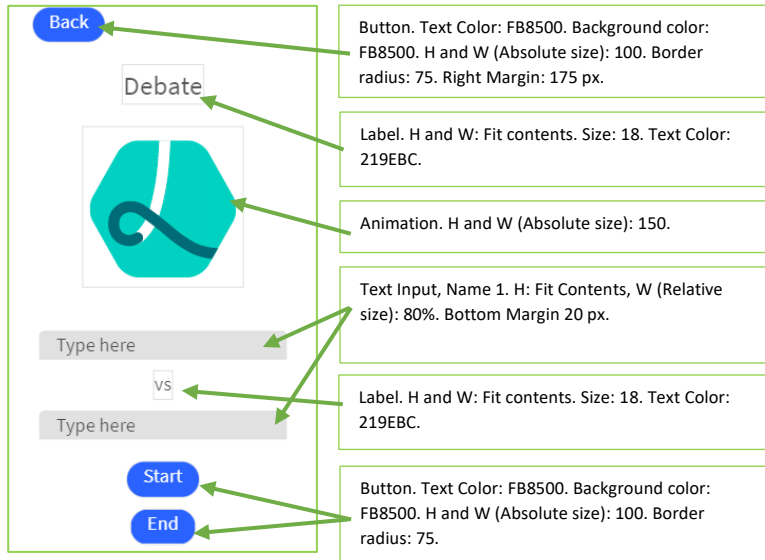
Response Time of Elements upon starting	Robotium	Click on Mobile Application Icon	0.5 Seconds
Debater 1 name, Debater 2 name.	Observation	Log In using trainee data; Email Id: studentx@school.com Password: 1987 Start debate using trainer account; Email Id: studentz@school.com Password: 1989	Shows current Debater 1 and 2 names set by Trainer in debate settings screen.
Debating Member Splash Screen	Observation	Log In using trainee data; Email Id: studentx@school.com Password: 1987 Start 'Student X' as debater 1 using trainer account; Email Id: studentz@school.com Password: 1989	Navigate to Splash Screen
Debater 1 like button	Observation	Click on Debater 1 like button	Disable button for 15 seconds and then enable.
Debater 2 like button	Observation	Click on Debater 2 like button	Disable button for 15 seconds and then enable.
Debate Ended by Trainer	Observation	Log In using trainee data; Email Id: studentx@school.com Password: 1987 End debate using trainer account; Email Id: studentz@school.com Password: 1989	Navigate to Trainee Home Page

Strand ii. Develops a range of feasible design ideas, using an appropriate medium(s) and detailed annotation, which can be correctly interpreted by other

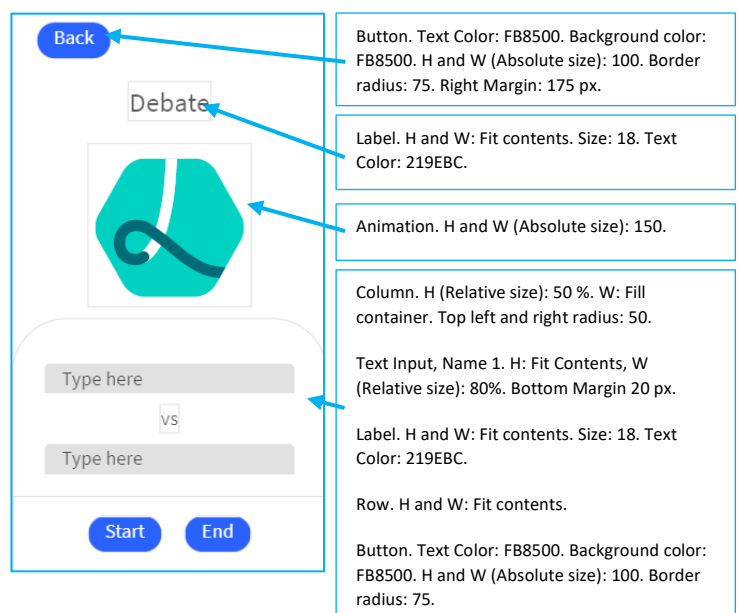
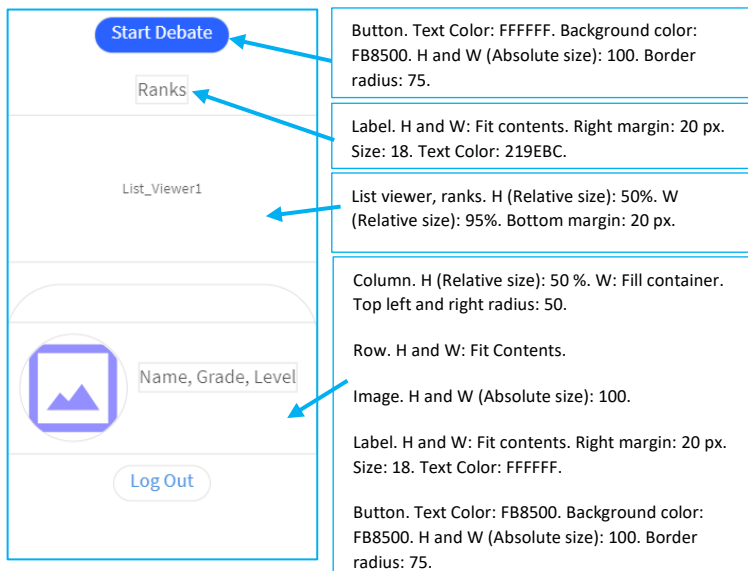
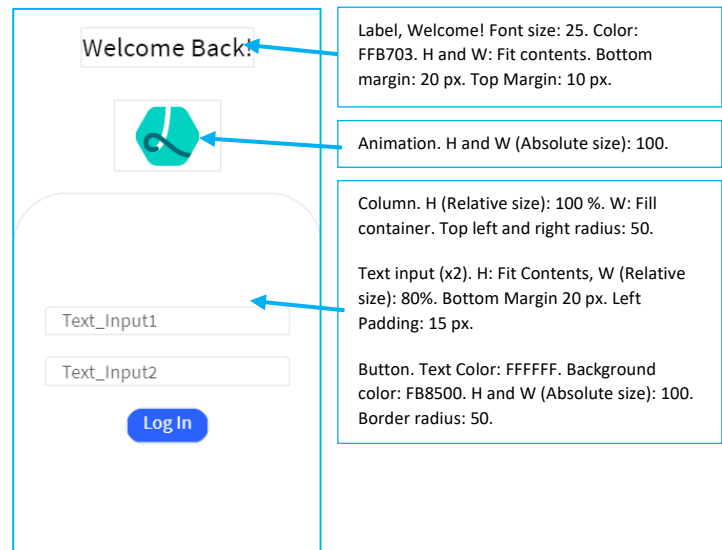
Note: Refer to *Navigation Map* in Appendix B.2

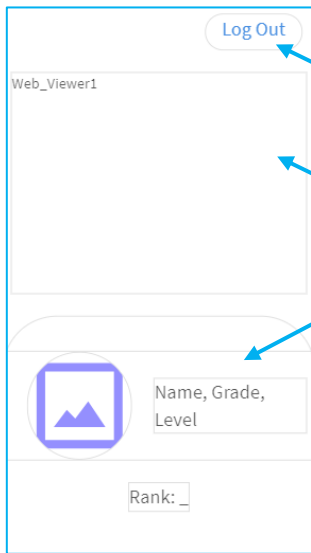
Design 1





Design 2





Button. Text Color: FFFFFFF. Background color: FB8500. H and W (Absolute size): 100. Border radius: 75. Left Margin: 125 px. Bottom Margin: 20 px.

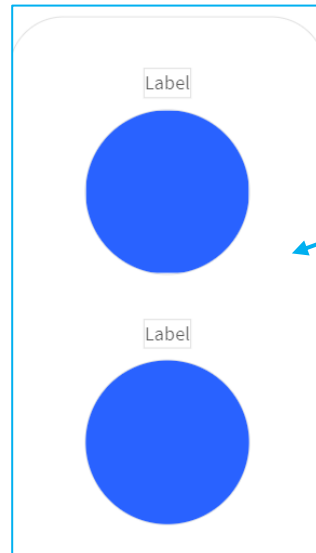
Web Viewer, graph. H (Relative size): 50%. W (Relative size): 95%. Bottom margin: 20 px.

Column. H: Fit contents. W: Fill container. Top left and right radius: 50.

Image. H and W (Absolute size): 100.

Label. H and W: Fit contents. Right margin: 20 px. Size: 18. Text Color: FFFFFFF.

Label. H and W: Fit contents. Right margin: 20 px. Size: 18. Text Color: 219EBC.



Column. H: Fit contents. W: Fill container. Top left and right radius: 50.

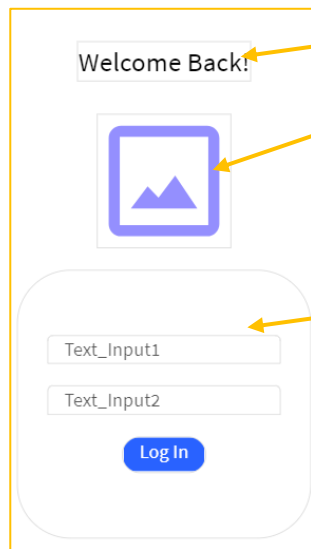
Label. H and W: Fit contents. Size: 18. Text Color: 219EBC. Bottom margin: 20 px.

Button. Text Color: FFFFFFF. Background color: FB8500. H and W (Absolute size): 100. Border radius: 75.

Label. H and W: Fit contents. Size: 18. Text Color: 219EBC. Bottom margin: 20 px.

Button. Text Color: FFFFFFF. Background color: FB8500. H and W (Absolute size): 100. Border radius: 75.

Design 3



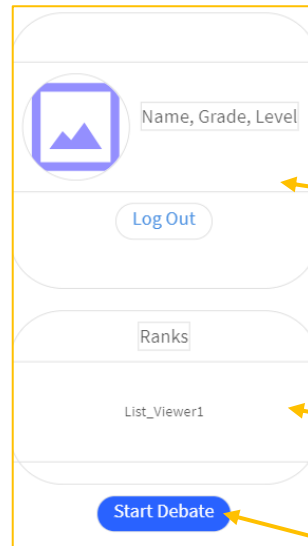
Label, Welcome! Font size: 25. Color: FFB703. H and W: Fit contents. Bottom margin: 20 px.

Image, Logo. H and W (Absolute Size): 125. Bottom margin: 15.

Column. H (Relative size): 100 %. W: Fill container. Border Radius: 50.

Text input (x2). H: Fit Contents, W (Relative size): 80%. Bottom Margin 20 px. Left Padding: 15 px.

Button. Text Color: FFFFFFF. Background color: FB8500. H and W (Absolute size): 100. Border radius: 50.



Column. H (Relative size): 50 %. W: Fill container. Border radius: 50.

Row. H and W: Fit Contents.

Image. H and W (Absolute size): 100.

Label. H and W: Fit contents. Right margin: 20 px. Size: 18. Text Color: FFFFFFF.

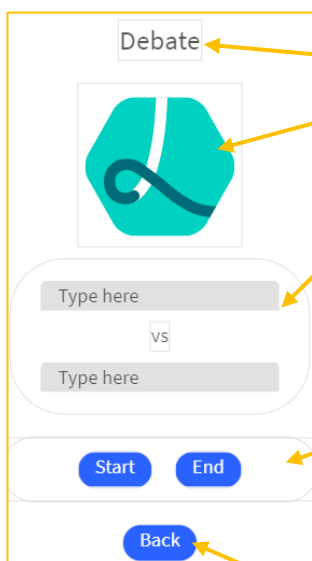
Button. Text Color: FB8500. Background color: FB8500. H and W (Absolute size): 100. Border radius: 75.

Column. H (Relative size): 50 %. W: Fill container. Border radius: 50.

Label. H and W: Fit contents. Right margin: 20 px. Size: 18. Text Color: 219EBC.

List viewer, ranks. H (Relative size): 50%. W (Relative size): 95%. Bottom margin: 20 px.

Button. Text Color: FFFFFFF. Background color: FB8500. H and W (Absolute size): 100. Border radius: 75.



Label. H and W: Fit contents. Size: 18. Text Color: 219EBC.

Animation. H and W (Absolute size): 150.

Column. H (Relative size): 50 %. W: Fill container. Top left and right radius: 50.

Text Input, Name 1. H: Fit Contents, W (Relative size): 80%. Bottom Margin 20 px.

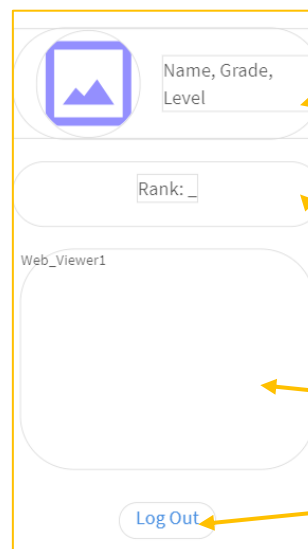
Label. H and W: Fit contents. Size: 18. Text Color: 219EBC.

Column. H (Relative size): 50 %. W: Fill container. Top left and right radius: 50.

Row. H and W: Fit contents.

Button. Text Color: FB8500. Background color: FB8500. H and W (Absolute size): 100. Border radius: 75.

Button. Text Color: FB8500. Background color: FB8500. H and W (Absolute size): 100. Border radius: 75.



Column. H: Fit contents. W: Fill container. Top left and right radius: 50. Bottom margin: 20 px.

Image. H and W (Absolute size): 100.

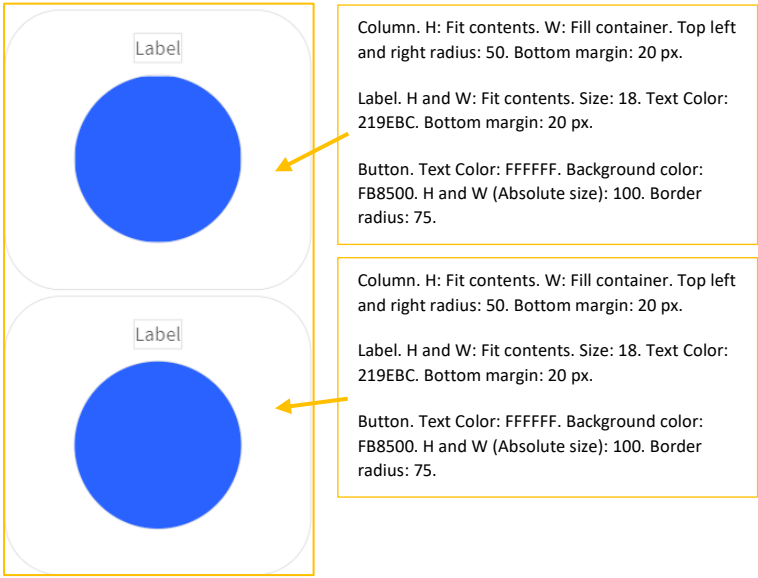
Label. H and W: Fit contents. Right margin: 20 px. Size: 18. Text Color: FFFFFFF.

Column. H: Fit contents. W: Fill container. Top left and right radius: 50. Bottom margin: 20 px.

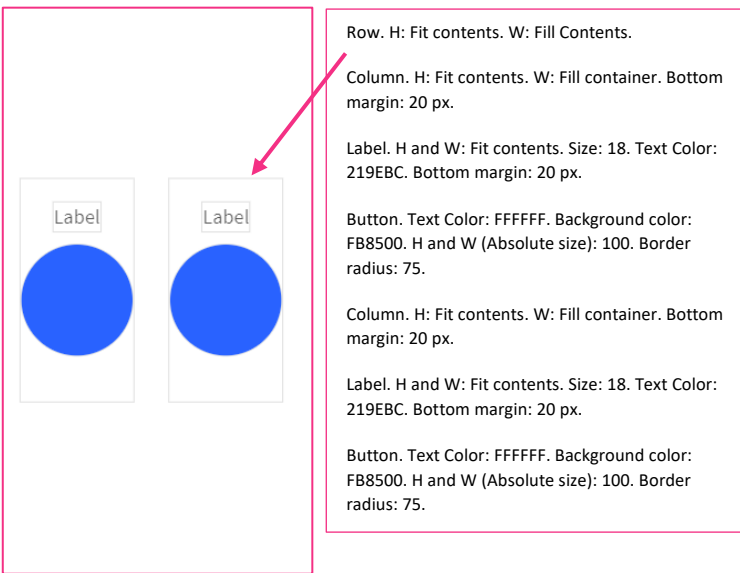
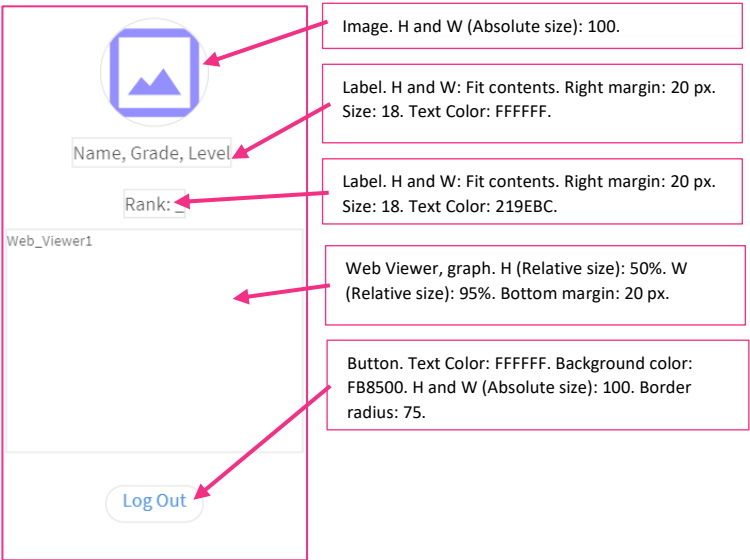
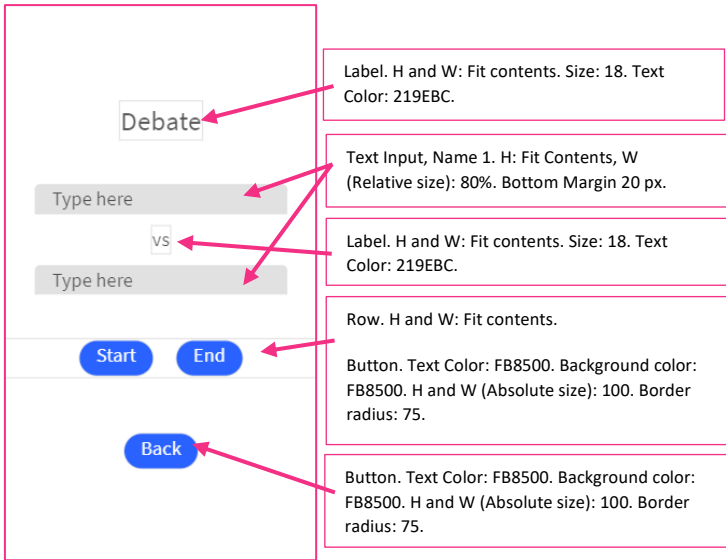
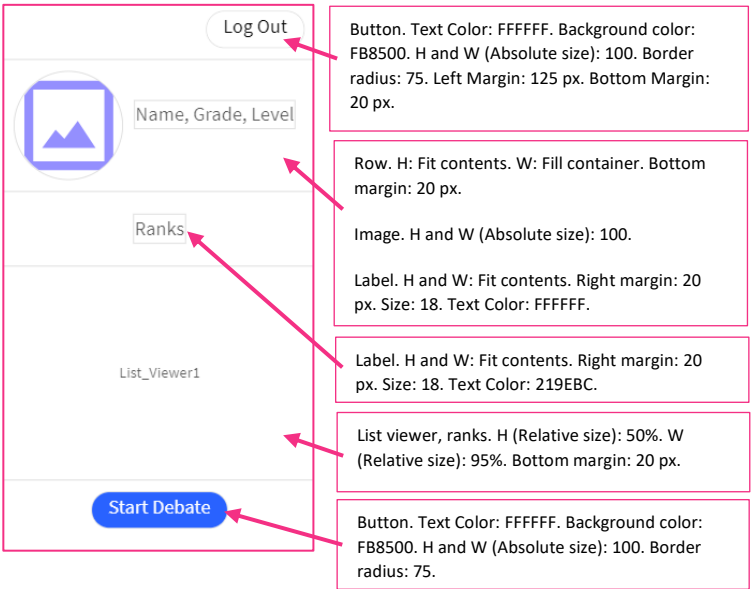
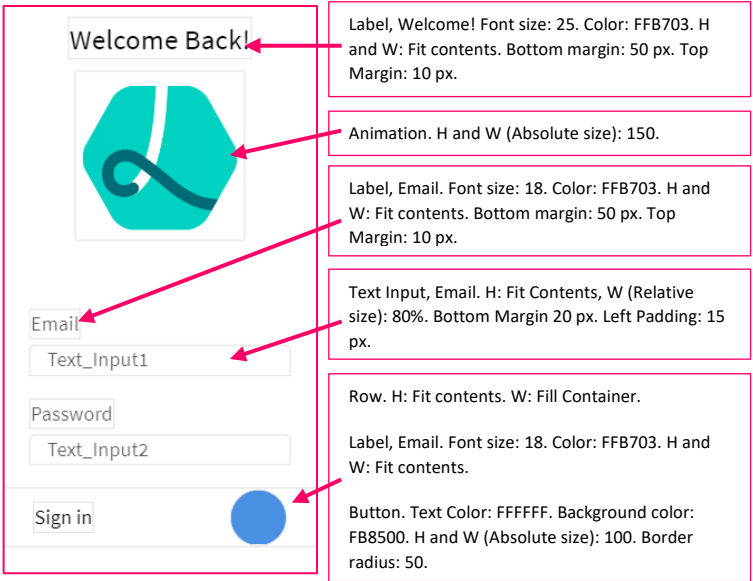
Label. H and W: Fit contents. Right margin: 20 px. Size: 18. Text Color: 219EBC.

Web Viewer, graph. H (Relative size): 50%. W (Relative size): 95%. Bottom margin: 20 px. Border Radius: 50.

Button. Text Color: FFFFFFF. Background color: FB8500. H and W (Absolute size): 100. Border radius: 75.



Design 4



Strand iii. Presents the chosen design and justifies fully and critically its selection with detailed reference to the design specification

Note: Score given against design success criterion (**Refer to Appendix B.1**) and is the average of the data collected. Margin of error: Rounded to 2 decimal points.

(Refer to Appendix B.3.1)

Criteria	Design 1	Design 2	Design 3	Design 4
Color Scheme (x/5)	3.43	4.29	4.43	2.71
Fonts and sizes (x/3)	2.14	2.43	2.86	1.71
Minimalistic and modern (curved edges) (x/1)	0.57	0.86	1	0.29
Total (x/9)	6.14	7.589	8.29	4.71

(Refer to Appendix B.3.2)

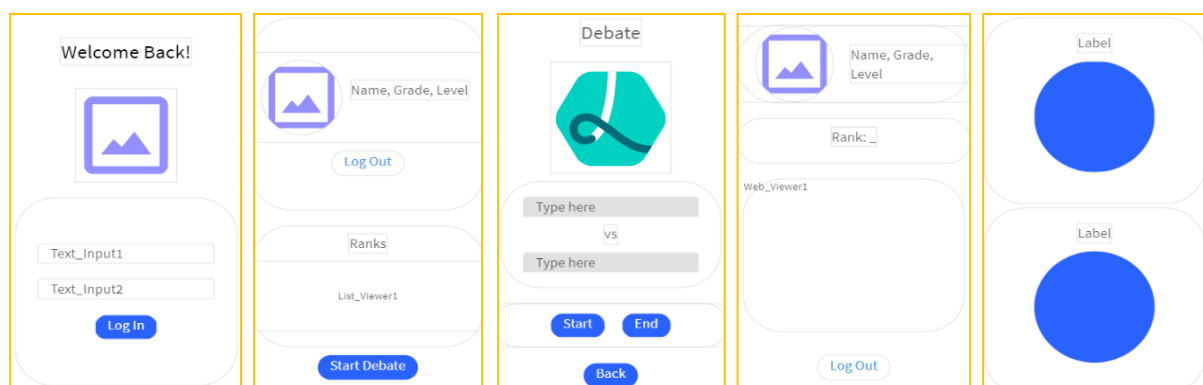
Client Feedback: My client chose design 3 as the most visually appealing from all the design as it used rounded elements that are common to today's graphic user interphases. In addition, she liked the use of the color scheme within the app elements. This prevented it from distracting the user and looking cluttered. Lastly, she liked the use of animations which made the application appear more dynamic and visually engaging.

Peer Feedback: Through peer feedback, I was able to understand the application was aesthetically pleasing due to the organization of elements such as the buttons which are located at the bottom of the screen. Thus, allowing easy access for the user's fingers. In addition, the navigation through the application appeared simple and clear for an MUN member. The application, also provides a modern appeal due to the curvature of the elements that match the design of today's smartphones.

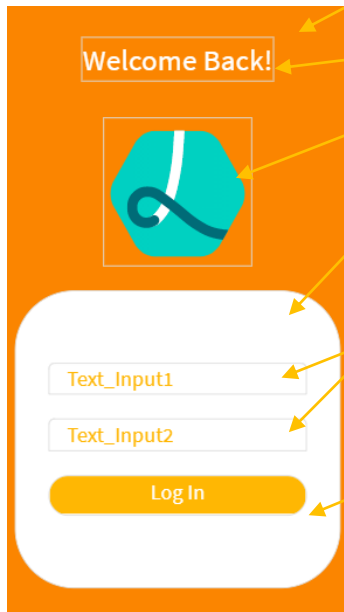
Design Specification: The chosen design satisfies the design specification by meeting the required functionality and utilizing the set font and font size (**Refer to Criterion B; Design Specification, A.1**). Along with the chosen design scheme (**Refer to Criterion B; Design Specification, A.2**). In addition, it is minimalistic and modern (**Refer to Criterion B; Design Specification, A.3**), thus completing the requirements.

Design Objective	Chosen Design
User	Caters to both trainer and trainee users.
Function and performance	Simple navigation, uncluttered design, smooth transitions.
Developer Requirements	Thunkable, HTML, API's, Database Querying
Aesthetics	Use of color scheme, font and their sizes, custom animations.
Time constraints and relation to the project theme	The product has a limitation of time of 1 month at maximum. It relates to the global context and exploration.

Chosen Design: Design 3



Strand iv. Develop accurate and detailed planning drawings/diagrams and outline the requirements for the creation of the chosen solution.



Background Color: FFB703.

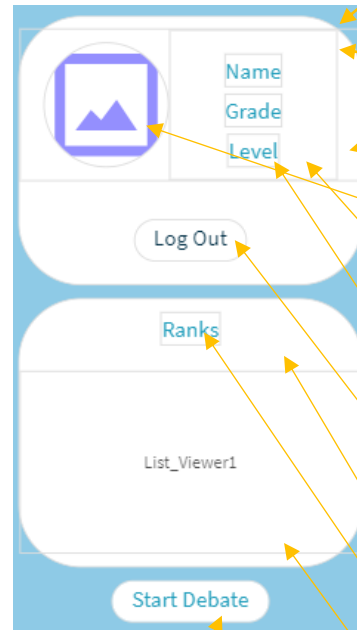
Label, Welcome Back! Font Size: 25. Color: FFFF. H and W: Fit Contents. Top Margin: 10 px. Bottom Margin: 15 px.

Animation. H and W: Absolute size, 200.

Column. H: Absolute size, 250. W: Relative size, 95%. Background Color: FFFFFFFF. Top margin: 20 px. Bottom Margin: 10 px. Border Radius: 50.

Text Input, Email. Keyboard Type: Email Address. Text color, FFB703. Font size, 18. H: Fit Contents. W: Relative Size, 80%. Bottom Margin: 20 px. Left Padding: 15 px. Border Color: FFB703. (Repeat for Text Input 2, Password)

Button, Log in. Text Color: FFFFFFFF. Font size: 18. Background Color: FFB703. H: Fit Container. W: Relative size, 80%. Border Radius: 75.



Background Color: 8ECAE6.

Column. H: Fill Container. W: Relative size, 95%. Background Color: FFFFFFFF. Top margin: 10 px. Bottom Margin: 10 px. Border Radius: 50.

Row. H and W: Fit Container. Bottom margin: 20 px.

Image. H and W: Absolute size, 100 px. Top and Bottom margin: 10 px. Border Radius: 50.

Column. H: Fill Container. W: Relative size, 50 %.

Label, Name. Text Color: 219EBC. Font size: 18. H and W: Fit Contents. Top Margin: 10 px. Bottom Margin: 5 px. (Repeat for Grade and Level Text Input)

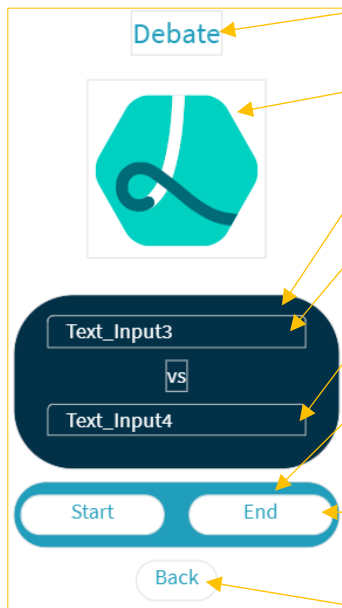
Button, Log out. Text Color: 023047. Font size: 18. H and W: Fit Contents. Bottom and Top Margin: 10 px.

Column. H: Fill Container. W: Relative size, 95%. Background Color: FFFFFFFF. Border Radius: 50.

Label, Name. Text Color: 219EBC. Font size: 18. H and W: Fit Contents. Top Margin: 10 px. Bottom Margin: 20 px.

List Viewer. H: Fill container. Bottom Margin: 10 px.

Button, Log out. Text Color: 219EBC. Background color: FFFFFFFF. Font size: 18. H and W: Fit Contents. Bottom and Top Margin: 10 px.



Label. H and W: Fit contents. Size: 18. Text Color: 219EBC.

Animation. H and W (Absolute size): 150.

Column. H (Relative size): 50 %. W: Fill container. Top left and right radius: 50.

Text Input, Name 1. H: Fit Contents, W (Relative size): 80%. Bottom Margin 20 px.

Label. H and W: Fit contents. Size: 18. Text Color: 219EBC.

Column. H (Relative size): 50 %. W: Fill container. Top left and right radius: 50.

Row. H and W: Fit contents.

Button. Text Color: FB8500. Background color: FB8500. H and W (Absolute size): 100. Border radius: 75.

Button. Text Color: FB8500. Background color: FB8500. H and W (Absolute size): 100. Border radius: 75.

Column. H: Fit contents. W: Fill container. Top left and right radius: 50. Bottom margin: 20 px.

Image. H and W (Absolute size): 100.

Label. H and W: Fit contents. Right margin: 20 px. Size: 18. Text Color: FFFFFFFF.

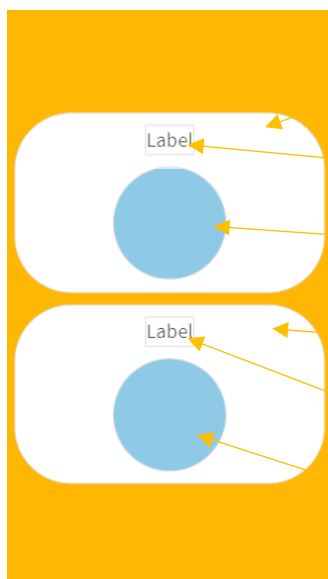
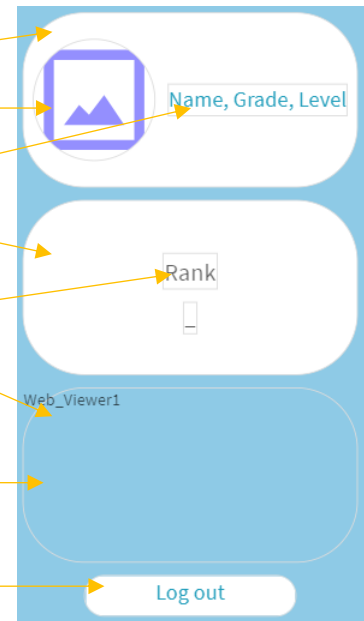
Column. H: Fit contents. W: Fill container. Top left and right radius: 50. Bottom margin: 20 px.

Label. H and W: Fit contents. Right margin: 20 px. Size: 18. Text Color: 219EBC.

Column. H: Fit contents. W: Fill container. Top left and right radius: 50. Bottom margin: 20 px.

Web Viewer, graph. H (Relative size): 50%. W (Relative size): 95%. Bottom margin: 20 px. Border Radius: 50.

Button. Text Color: FFFFFFFF. Background color: FB8500. H and W (Absolute size): 100. Border radius: 75.



Column. H: Fit contents. W: Fill container. Top left and right radius: 50. Bottom margin: 20 px.

Label. H and W: Fit contents. Size: 18. Text Color: 219EBC. Bottom margin: 20 px.

Button. Text Color: FFFFFFFF. Background color: FB8500. H and W (Absolute size): 100. Border radius: 75.

Column. H: Fit contents. W: Fill container. Top left and right radius: 50. Bottom margin: 20 px.

Label. H and W: Fit contents. Size: 18. Text Color: 219EBC. Bottom margin: 20 px.

Button. Text Color: FFFFFFFF. Background color: FB8500. H and W (Absolute size): 100. Border radius: 75.

Requirements (Developer Side):

General	A Laptop, Internet, A browser, Google Account, Thinkable, Airtable, Firebase.
Design	1. Visible Components. (Examples include buttons, loading icons, and images) 2. Invisible Components. (Examples include sounds, timers, and databases)
Division of Components	1. User Interphase: They are visible components that users can move, click on, and type in. 2. Layout: Helps set up and organize your app. 3. Voice: Consists of AI features such as understanding speech to translation and image recognition. 4. Image: It contains components to upload pictures to your user's device and to the cloud. It also has components for displaying pictures and animations 5. Data: Components for collecting, storing and, retrieving data in your app. 6. Location: Has a location sensor and maps by google and apple 7. Authentication: It enables your app to verify if something is true, like a password. 8. Organization of Components (Component tree): It contains a list of the visible and invisible components in your app. You can move components around on the component tree to change the location of a component in your app.
Blocks	Every component has its own set of blocks to start or trigger an event and set and change properties. Division of Blocks 1. Control: The "this, do that" block is commonly used with Logic blocks that define the conditions that lead one event to the next. 2. Logic: Has statement and value blocks used with Control blocks. 3. Math: Used to perform calculations and introduce randomness. Ex.  4. Text: Has a text value, reformat text and analyse text feature 5. Lists: Allows you to create a list, sort list, select from a list, analyse list, modify list and shuffle list. 6. Colour: Allows you to set RGBA colour, set RGB colour and set HSV colour 7. Objects: It pairs well with the Web API, Map and Location Sensor components which often send data in object format. 8. Device: Allows you to get current time and date, get mobile operating system, get online/offline status, set device to vibrate and dismiss keyboard. 9. Functions: Allows you to create App, stored and cloud variables.

Criterion C

Strand i. Constructs a detailed and logical plan, which describes the efficient use of time and resources, sufficient for peers to be able to

Note: The application code is dependent upon the user interphase, therefore the UI needs to be built before.

Timeline established is efficient as there is a buffer period established for small errors.

Short Term Goals →

1. Creating a Thunkable X Project – Jan 5th, 2021 (30 minutes)

Resources: Thunkable X, Gmail Account

Sign into Thunkable using a Gmail ID and click on the 'create new app' button. Click on the App Settings from the top panel. Rename the app to MUN Club and upload the app logo as the Universal Model UN Logo.

2. Creating Log in Page UI – Jan 6th, 2021 (1 Hour)

Resource: Thunkable X Visible Components (Labels, Columns, Text Inputs, Buttons, Images)

Rename the screen to 'Log In'. Drag and drop the following visible components from the 'Add Components' section in the following order on the screen;

- Label (L: Welcome Back)
- Animation (A: 1)
- Column (C: 1)
 - Text Input (TI: Email)
 - Text Input (TI: Password)
 - Button (B: Log In)
 - Loading Icon (LI: 1)
 - Label (L: Error 1)

Note: Rename the components as written above.

Set the allocated colors, font sizes, border radii, margins, heights and widths according to the specifications set **(Refer to Criterion B; Strand iv.)** Go to a free to use animations website such as Lottie Files and download the chosen animation as JSON file. Upload the animation to the files in the top right panel and link the animation to the file in the right panel.

3. Creating Trainer Home Page UI – Jan 7th, 2021 (1 Hour)

Resources: Thunkable X Visible Components (Labels, Rows, Columns, Buttons, Images)

Add a new screen by clicking on the plus button in the top panel next to the screen 'Log In'. Rename the screen to 'Trainer Home'. Drag and drop the following visible components from the 'Add Components' section in the following order on the screen;

- Column
- Row (R: 1)
 - Image (I: Trainer Photo)
- Column (C: 2)
 - Label (L: Name 1)
 - Label (L: Grade 1)
 - Label (L: Level 1)

Task	Duration
Creating a Thunkable X Project	Jan 5 th (30 minutes)
Creating Log In Page UI	Jan 6 th (1 hour)
Creating Trainer Home Page UI	Jan 7 th (1 hour)
Creating Trainee Home Page UI	Jan 8 th (1 hour)
Creating Debate Settings Page UI	Jan 9 th (1 hour)
Creating KUDOS Page UI	Jan 10 th (1 hour)
Setting up Databases; Airtable and Firebase	Jan 11 th (2 hour)
Coding Log In Page	Jan 12 th (1 hour)
Coding Trainer Home Page	Jan 13 th (1 hour)
Coding Trainee Home Page	Jan 14 th (1 hour)
Coding Debate Settings Page	Jan 15 th (1 hour)
Coding KUDOS Page	Jan 16 th (1 hour)

- Button (B: Log Out 1)
- Column (C: 3)
 - Label (L: Rank 1)
 - List Viewer (LV: Ranking)
- Button (B: Start Debate)

Note: Rename the components as written above.

Set the allocated colors, font sizes, border radii, margins, heights and widths according to the specifications set **(Refer to Criterion B; Strand iv.)**

4. Creating Trainee Home Page UI – Jan 8th, 2021 (1 Hour)

Resources: Thunkable X Visible Components (Labels, Rows, Columns, Buttons, Images)

Add a new screen by clicking on the plus button in the top panel next to the screen 'Trainer Log In'. Rename the screen to 'Trainee Log In'. Drag and drop the following visible components from the 'Add Components' section in the following order on the screen;

- Row (R: 2)
 - Image (I: Trainee Image)
 - Column (C: 4)
 - Label (L: Name 2)
 - Label (L: Grade 2)
 - Label (L: Level 2)
- Column (C: 5)
 - Label (L: Rank 2)
 - Label (L: Rank 3)
- Web Viewer (WV: Graph)
- Button (B: Log Out 2)

Note: Rename the components as written above.

Set the allocated colors, font sizes, border radii, margins, heights and widths according to the specifications set **(Refer to Criterion B; Strand iv.)**

5. Creating Debate Settings Page UI – Jan 9th, 2021 (1 Hour)

Resources: Thunkable X Visible Components (Labels, Rows, Columns, Buttons, Images)

Add a new screen by clicking on the plus button in the top panel next to the screen 'Trainee Log In'. Rename the screen to 'Debate Settings'. Drag and drop the following visible components from the 'Add Components' section in the following order on the screen;

- Label (L: Debate)
- Animation (A: 2)
- Column (C: 6)
 - Button (B: Debater 1.1)
 - List Viewer (LV: Debater 1)
 - Label (L: vs)
 - Button (B: Debater 2.1)
 - List Viewer (LV: Debater 2)
- Row (R: 3)
 - Button (B: Start)
 - Button (B: End)
- Button (B: Back)

Note: Rename the components as written above.

Set the allocated colors, font sizes, border radii, margins, heights and widths according to the specifications set **(Refer to Criterion B; Strand iv.)** Go to a free to use animations website such as Lottie Files and download the chosen animation as JSON file. Upload the animation to the files in the top right panel and link the animation to the file in the right panel.

6. Creating KUDOS Page UI – Jan 10th, 2021 (1 Hour)

Resources: Thunkable X Visible Components (Labels, Rows, Columns, Buttons, Images)

Add a new screen by clicking on the plus button in the top panel next to the screen 'Debate Settings'. Rename the screen to 'KUDOS'. Drag and drop the following visible components from the 'Add Components' section in the following order on the screen;

- Column (C: 7)
 - Label (L: Debater 1)
 - Button (B: Debater 1.2)
- Column (C: 8)
 - Label (L: Debater 2)
 - Button (B: Debater 2.2)

Note: Rename the components as written above.

Set the allocated colors, font sizes, border radii, margins, heights and widths according to the specifications set **(Refer to Criterion B; Strand iv.)**

7. Settings up the Databases – Jan 11th, 2021 (2 Hours)

Resources: Thunkable X Invisible Components (Airtable), Airtable Account, Google Firebase, Gmail Account

Drag and drop the Airtable component on the screen. Then go to Airtable and sign up using a Gmail account. Then create a base from scratch and name it as 'MUN Base'. Under the base create 3 different tables in the following order; Trainee Log In, Trainer Log In, Debate.

For the Trainee Log In create the following columns in the following order;

- Email Id
- Password
- Photo
- Name
- Grade
- Level
- Points

For the Trainer Log In create the following columns in the following order;

- Email Id
- Password
- Photo
- Name
- Grade
- Level

Then go to the API documentation and get the API Key, Base ID, Table Name and View Name. After doing so, go back to Thunkable and click on the Airtable icon at the bottom of the web page and input all the data in the right panel.

Drag and drop the Firebase component on the screen. Then go to Firebase and sign up using a Gmail account. Then create a project and name it as 'MUN'. To add firebase to your web app, register it under the nick name, "MUN Club." Then link your API key and database URL to Thunkable. Then under real time database, set up a new database in test mode.

Then add a key, 'Start' with the value as "No".

8. Coding Log in Page – 12th Jan, 2021 (1 Hour)

Initialize app variable 'x' to an empty list.

Initialize cloud variable 'Row Number'.

When Log in Screen Starts, do

- Set app variable 'x' to empty list
- Set L: Error 1's visible to false
- Set LI: 1's visible to False

When B: Log in Click, do

- Set B: Log in's disabled to true
- Set App variable 'x' to empty list
- Set L: Error 1's visible to false
- Set LI: 1's visible to true
- Function (Sign In: Trainer)
- Function (Sign In: Trainee)

When TI: Email Click, do

- Set TI: Email's text to "(blank) "

9. Coding Trainer Home Page – 13th Jan, 2021 (1 Hour)

Initialize app variable 'raw data'

When Trainer Home Opens, do

- Call AT: Trainer Log In's get cell with row number as cloud variable row number and column name as 'photo', then do
 - Set app variable to get property URL of object get property 0 of object value.
 - Set I: Trainer Photo's picture to app variable raw data
- Call At: Trainer Log In's get cell with row number as cloud variable row number and column name as 'name', then do
 - Set L: Name's text to value
- Call At: Trainer Log In's get cell with row number as cloud variable row number and column name as 'grade', then do
 - Set L: Grade's text to value
- Call At: Trainer Log In's get cell with row number as cloud variable row number and column name as 'level', then do
 - Set L: Level's text to value

10. Coding Trainee Home Page – 14th Jan, 2021 (1 Hour)

Initialize app variable 'raw data 2'

When Trainer Home Opens, do

- Call RD: Has Debate Started's Add Listener with key Start
- Call RD: Has Debate Started's get with key Start, then do
 - If value is equal to yes, do
 - Navigate to KUDOS

- Call AT: Trainee Log In's get cell with row number as cloud variable row number and column name as 'photo', then do
 - Set app variable to get property URL of object get property 0 of object value.
 - Set I: Trainer Photo's picture to app variable raw data
- Call At: Trainee Log In's get cell with row number as cloud variable row number and column name as 'name', then do
 - Set L: Name's text to value
- Call At: Trainee Log In's get cell with row number as cloud variable row number and column name as 'grade', then do
 - Set L: Grade's text to value
- Call At: Trainee Log In's get cell with row number as cloud variable row number and column name as 'level', then do
 - Set L: Level's text to value

When B: Log Out 2 Click, do

- Navigate to Log in Screen

11. Coding Debate Settings Page – 15th Jan, 2021 (1 Hour)

Initialize app variable 'Debater 1 Name'.

Initialize app variable 'Debater 2 Name'.

Initialize app variable 'Debate row number'.

When Debate Settings Opens, do

- Set LV: D1's visible to false
- Set LV: D2's visible to false
- Set B: Debater 2's disabled to true
- Set B: Start's disabled to true

12. Coding Debate Page – 15th Jan, 2021 (1 Hour)

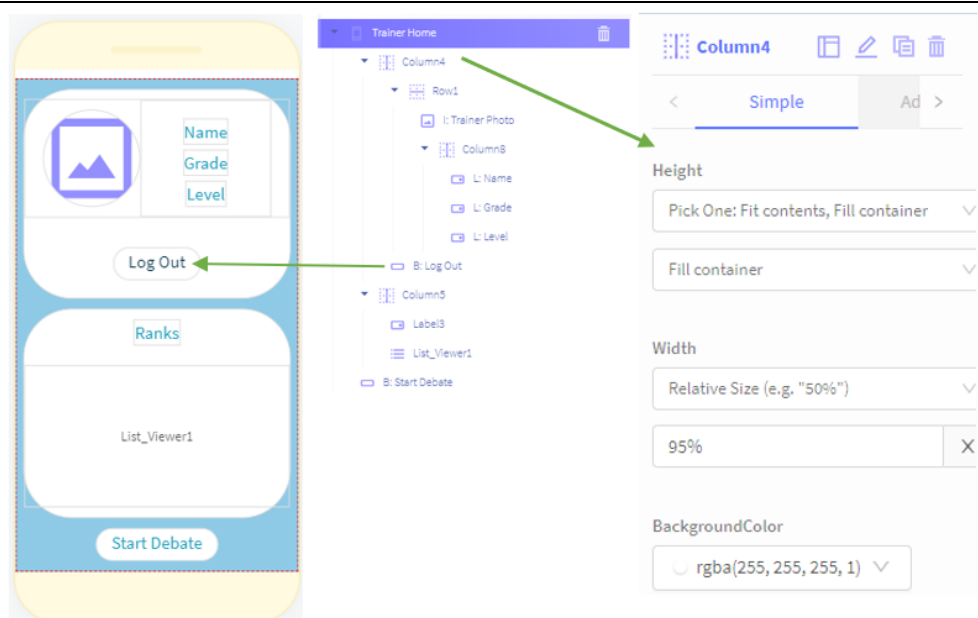
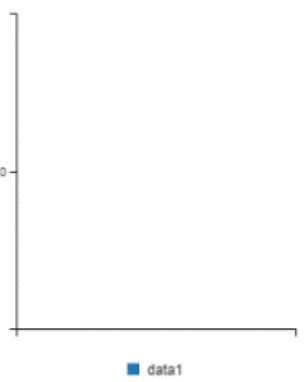
When KUDOS opens, do

- Set L: Debater 1's text to app variable Debater 1 name
- Set L: Debater 2's text to app variable Debater 2 name
- Set B: Debater 1's disabled to false
- Set B: Debater 2's disabled to false
- Call RD: Has debate started's Add listener with key 'Start', then do
 - If value is equal to 'Yes', do
 - Navigate to Trainee Home

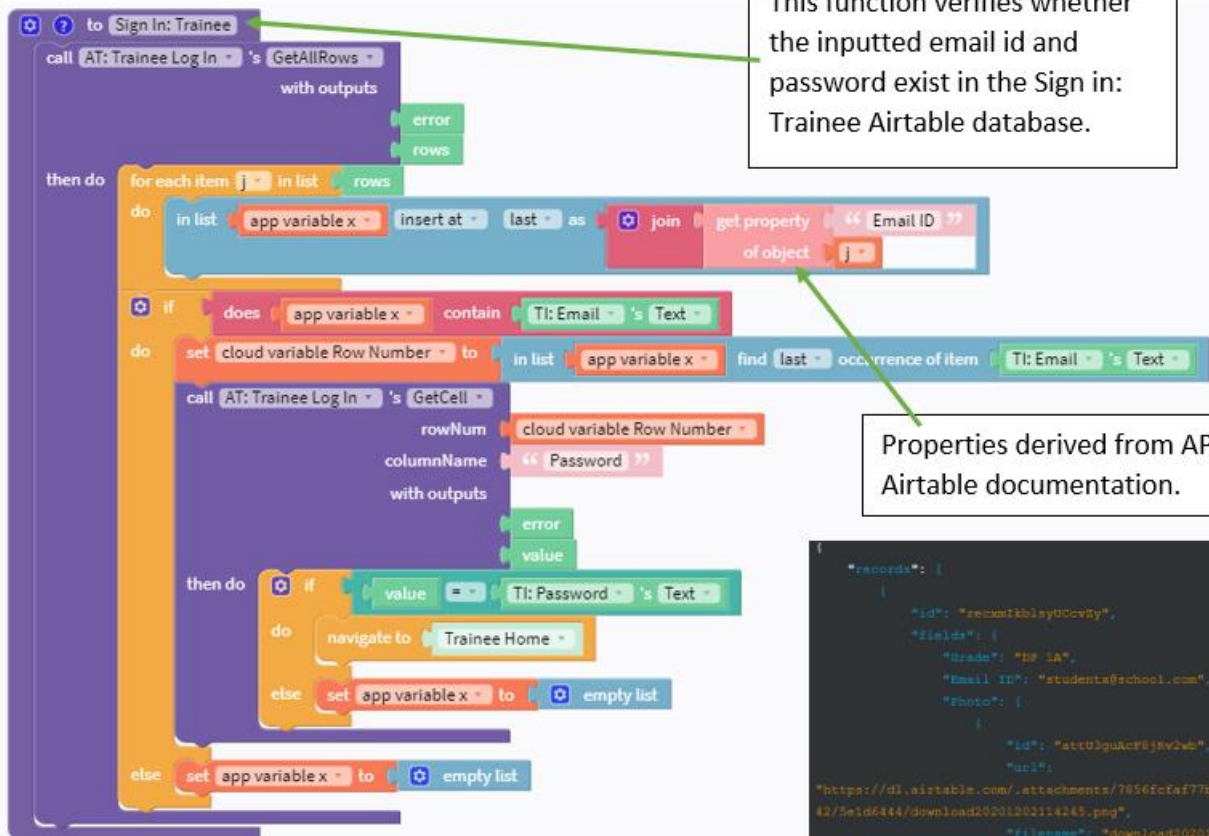
When RD: Has Debate Started Data Changed, do

- Call RD: Has Debate Started's Get with key 'Start', do
 - If value is equal to yes, do
 - Navigate to trainee home

Strand ii. Demonstrates excellent technical skills when making the solution.

Jan 7th, 2021	<i>Creating Trainer Page UI</i> To create the Trainer page, I followed the detailed instructions in Criterion C; Strand i which describe how to organize the elements, the exact numeric values for element descriptors such as the Height and Width of elements. The list viewer would display the student's name according to the ranks from highest to lowest. 
Jan 8th, 2021	<i>Creating Trainee Page UI</i> To create the Trainee Home Page, I followed the detailed instructions in Criterion C; Strand i which describe how to organize the elements, the exact numeric values for element descriptors such as the Height and Width of elements. For the web viewer, I used a sample of the following code in HTML for a web page which is then hosted on GitHub. 37 lines (36 sloc) 1.14 KB <pre> 1 <html> 2 <head> 3 <meta name="viewport" content="width=device-width, initial-scale=1"> 4 <script src="https://thunkable.github.io/webviewer-extension/thunkableWebviewerExtension.js" type="text/javascript"></script> 5 <link href="https://cdn.jsdelivr.net/npm/c3@0.7.15/c3.css" rel="stylesheet"> 6 <script src="https://d3js.org/d3.v5.min.js"></script> 7 <script src="https://cdn.jsdelivr.net/npm/c3@0.7.15/c3.min.js"></script> 8 </head> 9 <body> 10 <div id="chart"></div> 11 <script> 12 var chart = c3.generate({ 13 bindto: '#chart', 14 data: { 15 columns: [16 ['data1'], 17] 18 } 19 }); </pre> The graph looks like this on Thunkable; 
Jan 9th, 2021	<i>Creating Debate Settings Page UI</i> To create the Debate Page, I followed the detailed instructions in Criterion C; Strand i which describe how to organize the elements, the exact numeric values for element descriptors such as the Height and Width of elements. For the animation, I used a free-to-use JSON file from Lotte Files. This made it look more dynamic and interactive.
Jan 10th, 2021	<i>Creating KUDOS Page UI</i> To create the KUDOS Page, I followed the detailed instructions in Criterion C; Strand i which describe how to organize the elements, the exact numeric values for element descriptors such as the Height and Width of elements. I custom made a Clip Art image of a heart to match the color scheme set in Criterion B; strand I using Autodesk Sketchbook.

Jan 11th, 2021	<i>Setting up the Databases (Airtable and Firebase)</i> To setup the database, I needed a google account to sign in. For Airtable, I created one single base with 3 tables; Trainee log in, Trainer Log in and Debate. For Firebase, I created 1 project with a real-time database. The sole purpose of this was to identify where a debate had begun or not through a 'listener' in real time. This feature is not available on Airtable, which would have caused me to create a loop to keep checking for changes in data every 1 second. However, the listener made it easier to identify changes in data. To link the Firebase database to your Thunkable project, gather the following information;  <pre> <!-- The core Firebase JS SDK is always required and must be listed <script src="https://www.gstatic.com/firebasejs/8.2.9/firebase-app.js"> </script> <!-- TODO: Add SDKs for Firebase products that you want to use https://firebase.google.com/docs/web/setup#available-libraries <script src="https://www.gstatic.com/firebasejs/8.2.9/firebase-analytics"> </script> <script> // Your web app's Firebase configuration // For Firebase JS SDK v7.20.0 and later, measurementId is optional var firebaseConfig = { apiKey: "AIzaSyAeWYEIWei", authDomain: "mun-club.firebaseio.com", databaseURL: "https://mun-club-default-rtdb.firebaseio.com", projectId: "mun-club", storageBucket: "mun-club.appspot.com", messagingSenderId: "973573174892", appId: "1:973573174892:web:107bf3888e1ef28800a49", measurementId: "G-9H072CFBLH" }; // Initialize Firebase firebase.initializeApp(firebaseConfig); firebase.analytics(); </script> </pre>  . read <u>and</u> write: Describes if and when data is allowed to be read by users. <pre> { "rules": { ".read": "true", ".write": "true" } } </pre>
----------------------------------	---



```

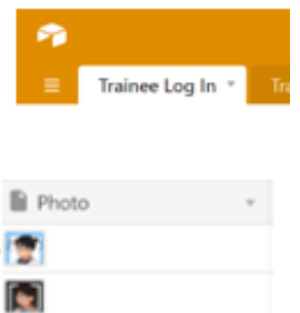
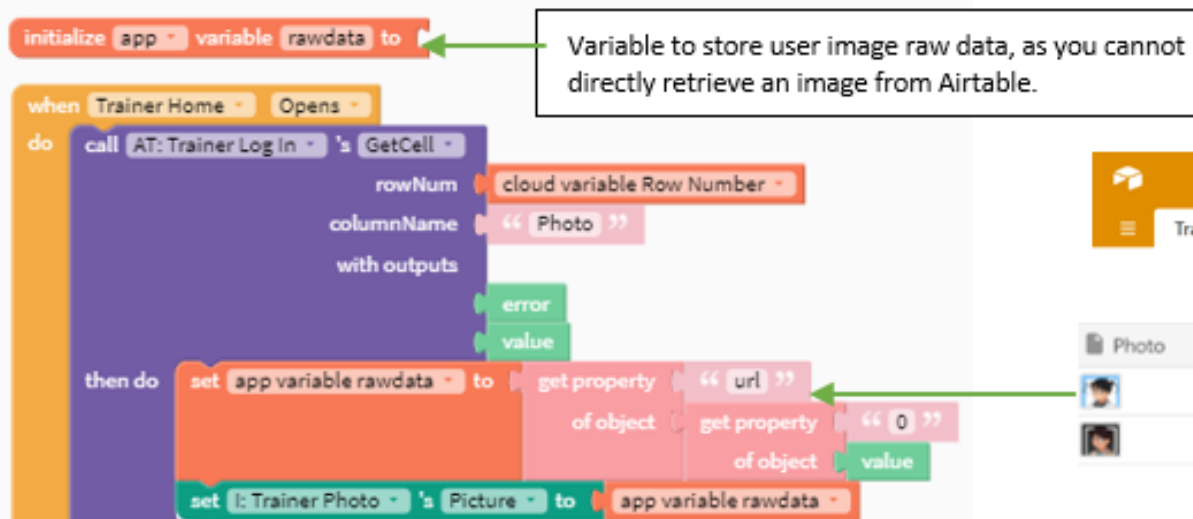
{
  "records": [
    {
      "id": "reemxIkblayU0Cov2y",
      "fields": {
        "Grade": "HS-1A",
        "Email ID": "students@school.com",
        "Photo": {
          "id": "attU3guAcF6jNv2wb",
          "url": "https://dl.airtable.com/.attachments/7856fcfaf77beb34e55437274740b042/5e1d6444/download20201202114245.png",
          "filename": "download20201202114245.png",
          "size": 300140,
          "type": "image/png",
          "thumbnails": {
            "small": {
              "url": "https://dl.airtable.com/.attachmentThumbnails/a15f19c9ad86c3efanade07da651e73d/764a94f0"
            }
          }
        }
      }
    }
  ]
}

```

Coding the Trainer Home Page

I followed the written description of code in **Criterion C; strand i**.

This page solely shows the trainer their personal information and the trainee ranking. It was extremely challenging to come up with appropriate code for organizing the ranks, therefore I took help from an expert; view on extreme right. I had to wait for two days until I got a response from the Thunkable expert, therefore extending the timeline by 2 days.





jared Power Thunker

Expert advice

Apr '20

I think a leader board would be great when stored like this

```
CLOUD listOfScores = [{(NAME:"user1",SCORE:123),(NAME:"user2",score:233)}
```

(so a list of objects)

then,

everytime you add a new highscore, instead of just adding to a list, you compare the new object property score to the all others starting at # 1, and if it's greater to that score insert that new object at postioin X.

when yo display the data. you display name and score next to each other in a list viewer but limit the result to the first 10 in the list

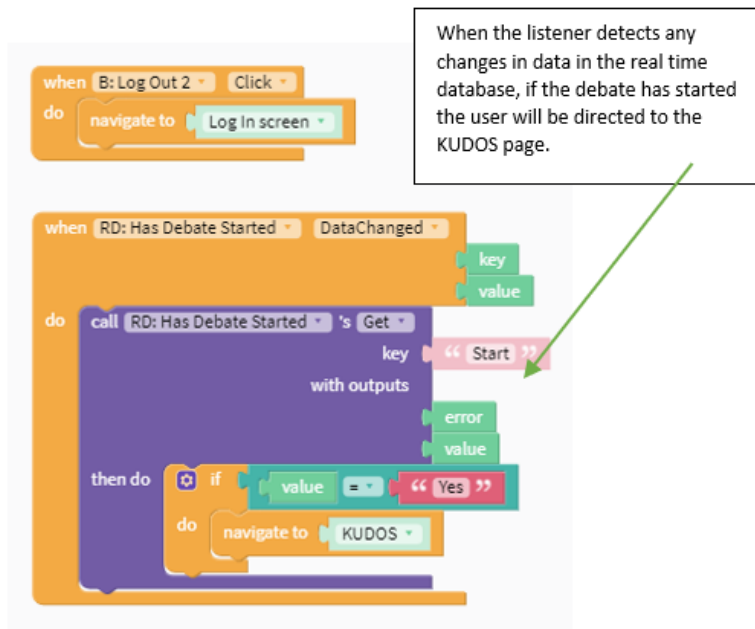
Jan
16th
and
17th,
2021

Coding the Trainee Home Page

I followed the written description of code in **Criterion C; strand i**.

Here as can be seen on the right, a lister is added. The trainee user cannot navigate to the KUDOS page themselves; the trainer needs to start a debate for that to happen.

Parts of the code were not saved when I exited on 16th in Thunkable and therefore, I had to re-organize the blocks on 17th.



Jan
18th,
2021

Coding the Debate Settings

I followed the written description of code in **Criterion C; strand i**.

Here the trainer can start a debate by choosing 2 trainees through a multiple-choice question.

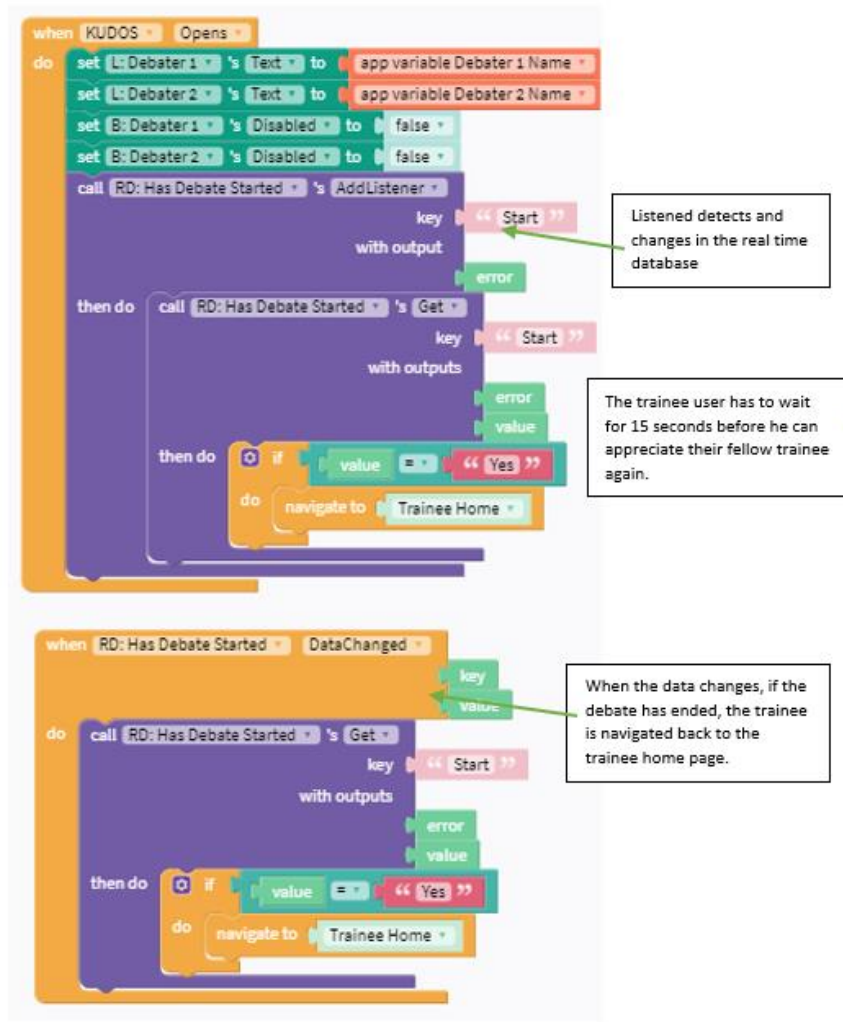
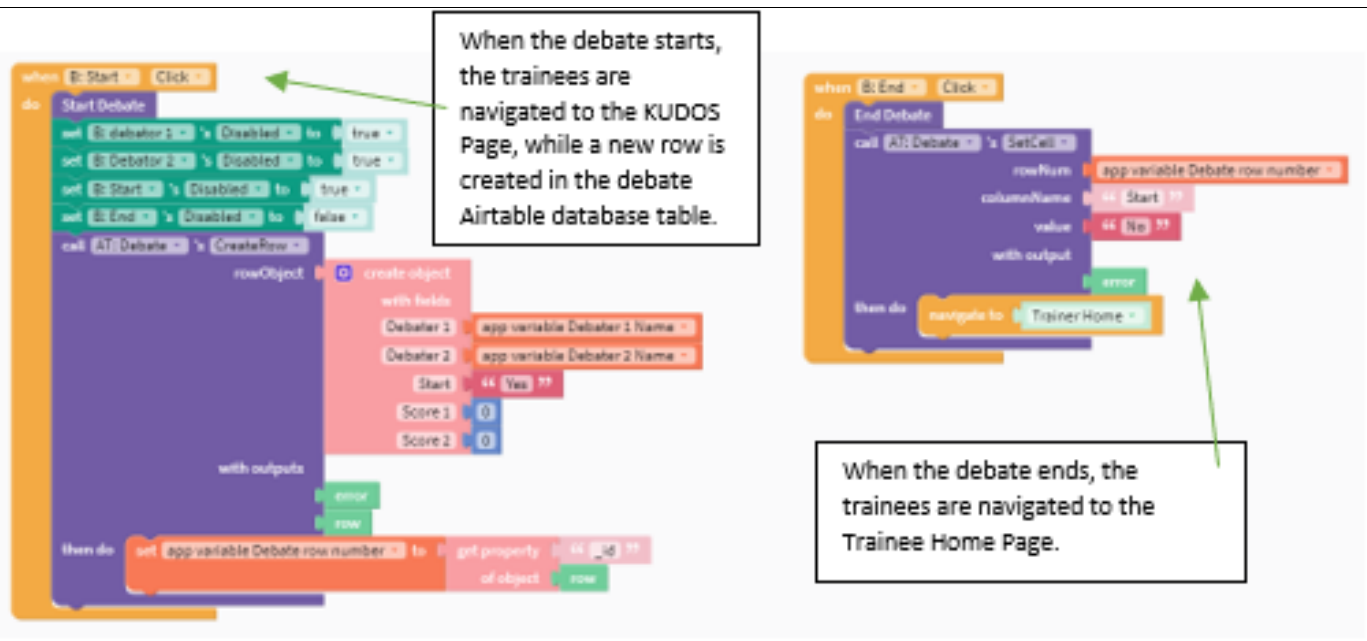
Jan 19th,

Coding the KUDOS

I followed the written description of code in **Criterion C; strand i.**

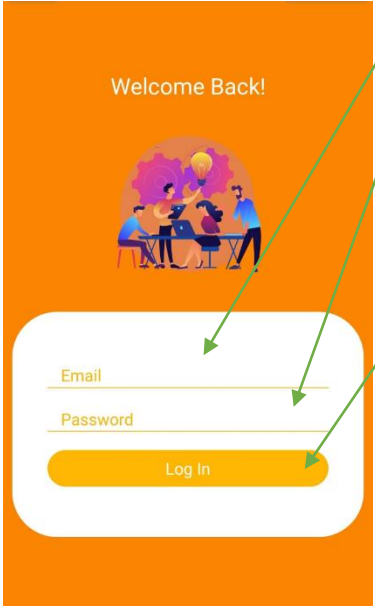
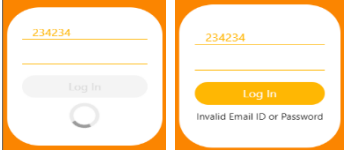
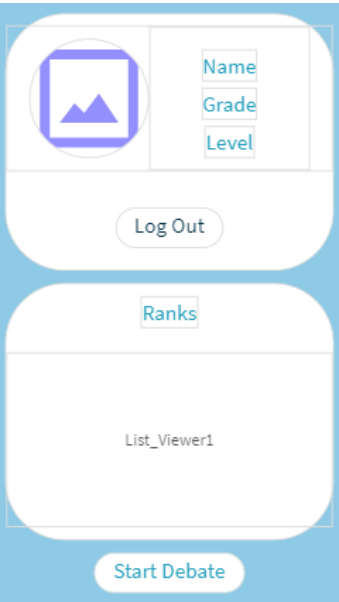
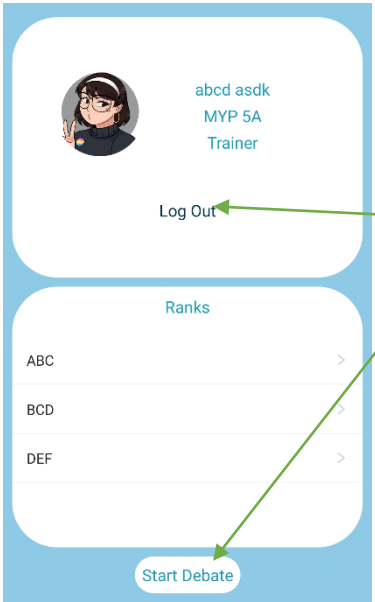
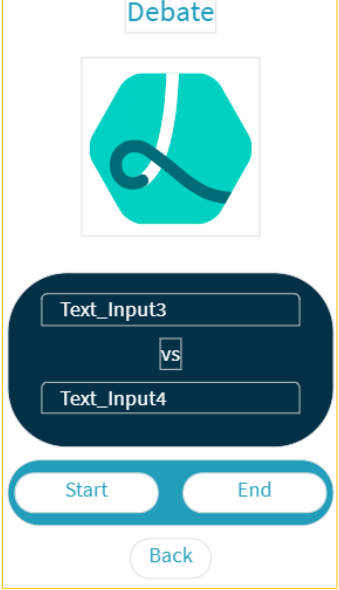
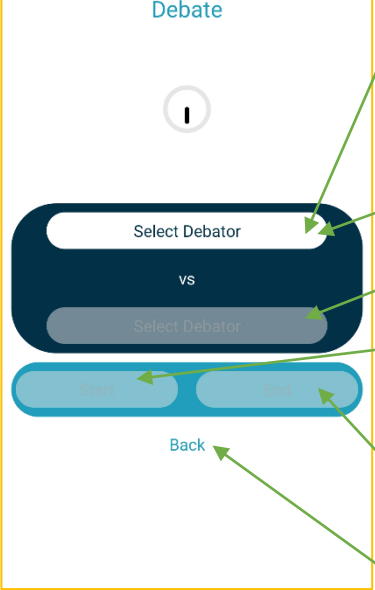
The 15 second wait interval, prevents a user from clicking the button from more than 20 times as;

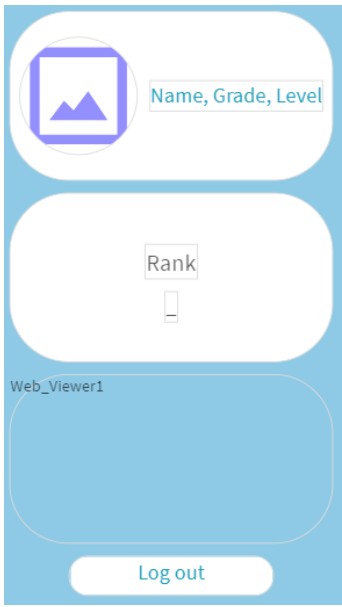
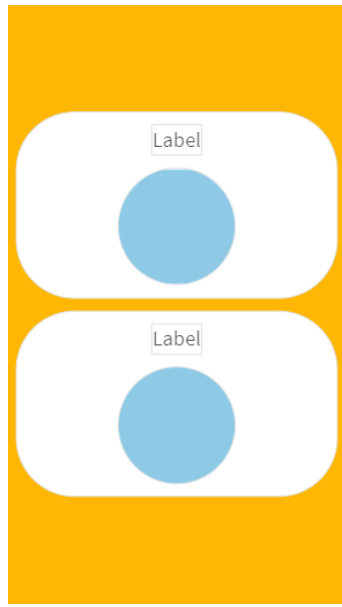
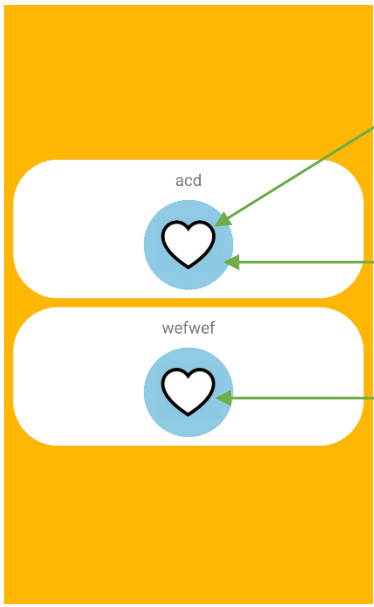
A debate typically lasts for 5 minutes or 300 seconds. Therefore, $\frac{300}{15} = 20$
A trainee can appreciate another member 20 time at maximum.



Strand iii. Follows the plan to create the solution, which functions as intended as is presented appropriately.

Criterion B Designs	Criterion C Product	Annotations
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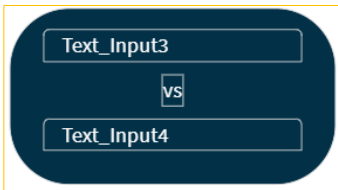
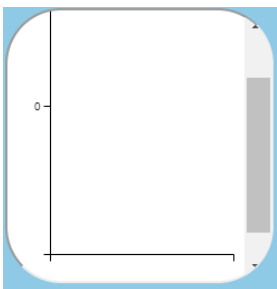
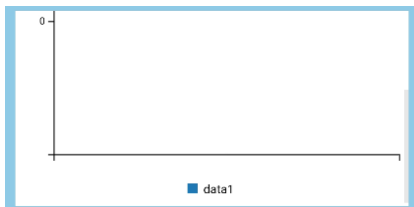
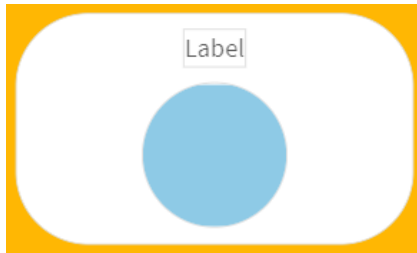
Log in Page			Functionality: Enter School Email Address Enter Password which is the student's school ID number (4-digit number) Authenticates Email address and password, then navigates trainers to Trainer Home Page and navigates trainees to the Trainee home page. If the authentication is invalid, an error message pops up. 
Trainer Home Page			Functionality: Returns user back to the Log in Page. Navigates user to the Debate Settings Page.
Debate Settings			Changes: List Viewer as a multiple-choice question instead of text input to choose debater. Functionality: Select name of debater 1 Select name of debater 2 Start Debate, navigates trainees to KUDOS page End Debate, navigates trainees to Trainee Home Page Return back to the Trainer home page.

Trainee Home Page	 Wireframe of the Trainee Home Page. It features a header with a profile icon and a label 'Name, Grade, Level'. Below this is a 'Rank' section with a placeholder for a rank value. At the bottom is a 'Log out' button. A 'Web_Viewer1' label is present near the bottom left.	 Final design of the Trainee Home Page. It features a header with a profile icon and text 'acd DP 1A Advanced'. Below this is a 'Rank' section showing 'Rank 1'. At the bottom is a 'Log out' button. A line graph is present in the middle section, labeled 'data1'. Changes: The Graph doesn't have curved edges as Thunkable doesn't allow this feature for Web Viewers. Functionality: Return back to the Log in Page.
KUDOS Page	 Wireframe of the KUDOS Page. It features two identical sections, each with a 'Label' and a blue circle.	 Final design of the KUDOS Page. It features two identical sections, each with a 'Label' and a heart icon. The first section is labeled 'acd' and the second is labeled 'wefwef'. Changes: The button background is heart as a plain solid color is too minimalistic and is misleading. Functionality: Increases the score of the first debater by 1 point. Increases the score of the second debater by 1 point.

Task	Goal Duration	Executed Duration
Creating a Thunkable X Project	Jan 5 th (30 min)	Jan 5 th (10 minutes)
Creating Log in Page UI	Jan 6 th (1 hour)	Jan 6 th (45 minutes)
Creating Trainer Home Page UI	Jan 7 th (1 hour)	Jan 7 th (1 hour)
Creating Trainee Home Page UI	Jan 8 th (30 min)	Jan 8 th (30 min)
Creating Debate Settings Page UI	Jan 9 th (1 hour)	Jan 9 th (1 hour)
Creating KUDOS Page UI	Jan 10 th (1 hour)	Jan 10 th (1 hour)
Setting up Databases; Airtable and Firebase	Jan 11 th (2 hours)	Jan 11 th (2 hours)
Coding Log in Page	Jan 12 th (1 hour)	Jan 12 th (1 hour)
Coding Trainer Home Page	Jan 13 th (1 hour)	Jan 13 th to 15 th (2 hours)
Coding Trainee Home Page	Jan 14 th (1 hour)	Jan 16 th and 17 th (1 hour + 30 min)
Coding Debate Settings Page	Jan 15 th (1 hour)	Jan 18 th (1 hour)
Coding KUDOS Page	Jan 16 th (1 hour)	Jan 19 th (1 hour)

Strand iv. Fully justifies changes made to the final chosen design and plan when making the solution.

My ultimate goal was the deliver a product that met all the requirements of my client to solve their problem of a lack of a sense of belonging, therefore I took certain liberties as a developer to modify the aesthetics;

Aspect	Original Design	Change in Final Product	Justification
Aesthetics			Screen: Debate Settings I chose a to put a list Viewer as a multiple-choice question instead of text input for the convenience of the user. Now the user simple has to tap on the trainee's name and start the debate, whereas in the original design the user would have to type out the full name of the trainee.
			Screen: Trainee Home Page I had intended for the graph to have a curved border to align all the elements with a similar theme and modern aesthetic. However, Thinkable does not permit this feature on Web Viewers.
			Screen: KUDOS Page The button background is heart as a plain solid color in the original design is too minimalistic and does not inform the user what the purpose of the button is.
Timeline	Coding Trainer Home Page: Jan 13 th (1 Hour)	Coding Trainer Home Page: Jan 13 th to 15 th (2 Hours)	I needed help from a Thinkable expert to create code for a ranking system. This exchange took around 2 days.
	Coding Trainer Home Page: Jan 13 th (1 Hour) Coding Trainee Home Page: 14 th (1 hour) Coding Debate Settings Page: 15 th (1 hour) Coding KUDOS Page: 16 th (1 hour)	Coding Trainer Home Page: Jan 13 th to 15 th (2 Hours) Coding Trainee Home Page: 16 th and 17 th (1 hour + 30 min) Coding Debate Settings Page: 18 th (1 hour) Coding KUDOS Page: 19 th (1 hour)	I had extended my set goals by 1 day as the trainee home page code had not saved on 16 th Jan. Therefore, I had to re-do block organization on 17 th Jan which extended my goals by a total of 3 days.

Criterion D

Strand i. design detailed and relevant testing methods, which generate data, to measure the success of the solution

User Trial (Refer to Appendix D.1.1)

In order to conduct the user trial, I did a 1-week trial of the mobile application with MUN Club members to determine its effects in its native environment. A questionnaire was then sent out at the end of the week.

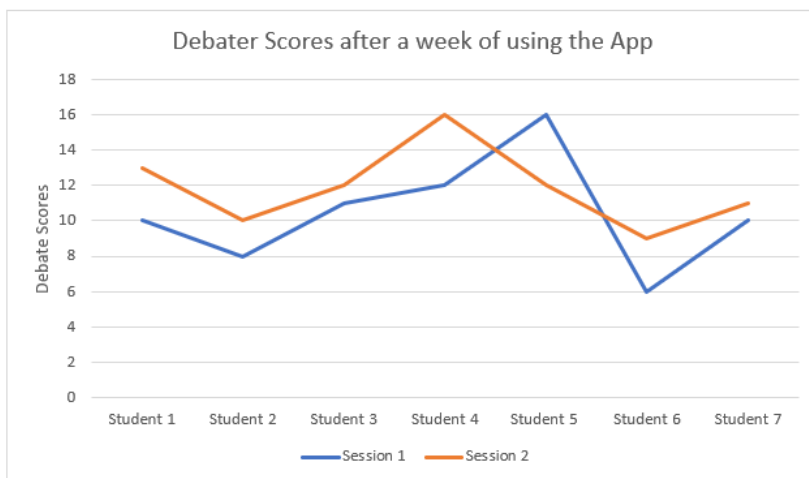
User Interphase: The use of the chosen color scheme in the design specification was given an average of approximately 8.43 out of 10. 100% of the sample users agreed that the font sizes were abided by and that the application was dynamic. The use of custom animations was given an average of approximately 8.43 out of 10 as well. The overall aesthetics were given an average of approximately 8.43 out of 10 as well.

Functionality: 100 % of the users were required to sign in using their school email id. 100% of the users were able to access all the pages. 100% of the sample users were also able to access the IOS and android version of the application. 71.4% of the sample users agree that the app promotes synergy and a sense of belonging in the club. The functional purpose of the application was given an average of 8.71 out of 10.

Improvements suggested: Including tips and tricks to improve debating and explanation of navigation and functionality.

Debate Scores:

Performance Testing (Refer to Success Criteria to view input, testing method and expected output)



General	
What is being tested	Actual Output
App Start-Up	0.783 Seconds, it passes .
Battery Time while using an app	Trainer: 3% Trainee: 2% (Margin of error, rounded off to whole number) Hence, it passes .
Memory Consumption	2.32% (Margin of error, rounded off to 3 significant figures) Hence, it passes .
Hardware/Software Variation	All devices reported no issue with the graphic in respect to the varying screen resolutions. and functioned as per mentioned in the design specification. Hence, it passes .
Log In Screen	
What is being tested	Actual Output
Log In screen Validity	1. Navigate to Trainer or Trainee Home Page. 2. Error Message 3. Error Message 4. Error Message 5. Error Message Hence, it passes .
Log In Screen Navigation	1. Navigates to Trainer Home Page 2. Navigates to Trainee Home Page Hence, it passes .
Response Time of Elements upon starting	0.746 seconds, it fails .
Trainer Home Page Screen	
What is being tested	Actual Output
Response Time of Elements upon starting	0.921 seconds, it fails .
User Image, Name, Grade, Level, Rank	Shows corresponding user image, name, grade, level and rank with email address from the Airtable. Hence, it passes .
Log Out button Navigation	Navigates to Log In Screen. Hence, it passes .

Start Debate Navigation	1. If a debate has not started; Navigates to Debate Settings Page 2. If a debate has been started by another trainer; Stays on Home Page Hence, it passes .
Debate Settings Page Screen	
What is being tested	Actual Output
Response Time of Elements upon starting	0.711 seconds, it fails .
Disabled buttons	None, it passes .
Start Debate Navigation	If a debate has been started by another trainer; Navigates to Home Page Hence, it passes .
Select Debater 1 Button	Shows List Viewer. Hence, it passes .
Select Debater 1 List Viewer	Hides List viewer and enables select debater 2 button. Hence, it passes .
Select Debater 2 Button	Shows List Viewer. Hence, it passes .
Select Debater 2 List Viewer	Hide List Viewer and enable Start Button. Hence, it passes .
Hide Debater 1 name from Select Debater 2 List Viewer	Not visible, it passes .
Start Button	Disables Start Button, enables end button. Hence, it passes .
End Button Button	Navigates to Home Screen Hence, it passes .
Back Button	Navigate to Home Screen Hence, it passes .
Trainee Home Page Screen	
What is being tested	Actual Output
Response Time of Elements upon starting	0.559 seconds, it fails .
User Image, Name, Grade, Level, Rank and graph	Shows corresponding user image, name, grade, level and rank with email address from the Airtable. Hence, it passes .
Log Out button Navigation	Navigates to Log In Screen. Hence, it passes .
Debate Started	Navigates to Home when debate begins. Hence, it passes .
KUDOS Screen	
What is being tested	Actual Output
Response Time of Elements upon starting	0.652 seconds, it fails .
Debater 1 name, Debater 2 name.	Shows current Debater 1 and 2 names set by Trainer in debate settings screen. Hence, it passes .
Debating Member Splash Screen	Navigates to Splash Screen. Hence, it passes .
Debater 1 like button	Disable button for 15 seconds and then enable. Hence, it passes .
Debater 2 like button	Disable button for 15 seconds and then enable. Hence, it passes .
Debate Ended by Trainer	Navigates to Trainee Home Page. Hence, it passes .

Expert Appraisal (Refer to Appendix D.1.2)

An .apk and .ipa file was sent to Ms. Mansi Suri, a computer engineer alongside the design specification.

Aesthetics: The mobile application successfully utilized the chosen color scheme, font, and font sizes. The use of a rounded border for all elements successfully provides it a 'modern' aesthetic. However, does not contribute to its minimalism. The use of animations makes it aesthetically dynamic, yet they are not MUN Specific.

Consumer: The application caters to both trainer and trainee consumers as there are pages dedicated to each type.

Safety and Size: Users are required to sign in with their school email id. A google pass can be used to make this process more efficient. In addition, a page should be added where the trainers can add more trainees in their database, rather than having to manually enter it in the database.

Functionality: The application has a common log-in screen for trainers and trainees. It then branches out further to a trainer home page and debate settings page, and trainee home page and KUDOS page. The application is also available on both IOS and android devices. It is important to note that the navigation can be confusing for a beginner, therefore some instructions should be given.

Product Comparison

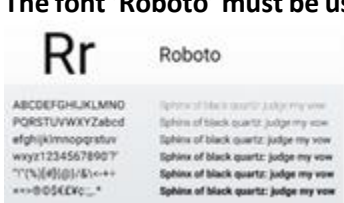
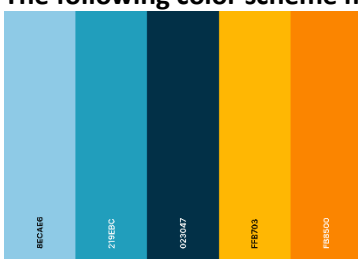
Why did I choose this product? I analyzed the following product in Criterion A while researching as it had similar functionality to my product.

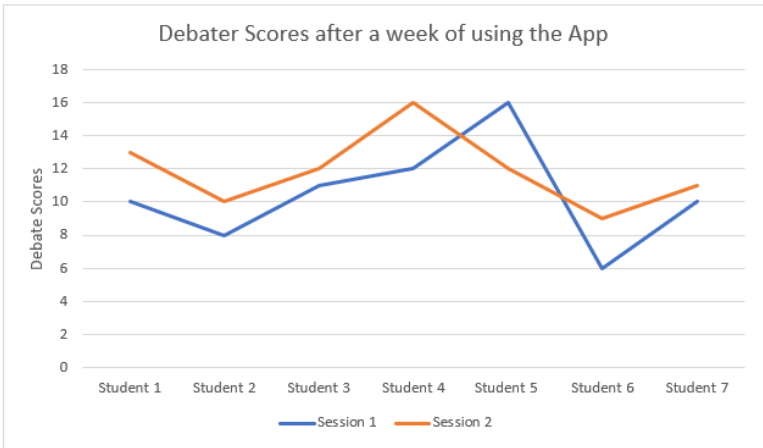
Note: Each aspect of the product is rated out of 10 to produce both quantitative and qualitative results

	MUN App (My Product)
Aesthetics: 9/10	My application uses a range of blue shades similar to the HR Cloud Employee app due to its benefits such as stimulating productivity and inspire truest which evoke a sense of belonging. In opposition to minimalism, my product is dynamic due to the custom animations.
Cost: 10/10	The mobile application is completely free.
Consumer: 9/10	The mobile application targets members of Model UN clubs and conferences.
Environment and Ergonomics: 8/10	The mobile application intends to evoke a sense of belonging that promotes synergy within the MUN Club. Similar to the HR Cloud it faces challenges in its navigation for new users.
Size and Safety: 8/10	Users are required to sign in with their school email id which verify them as members of the club. The application does not share its data with any 3 rd parties.
Functionality: 8/10	The mobile application allows trainees to appreciate their fellow debating trainees virtually that contributes to their lobbying score. This enables peer recognition as well and provides them a quantitatively value of where they rank in the club, without revealing it to other members.

Total Score for proposed solution: 52/60 vs. Total Score for HR Cloud: 44/50

Strand ii. critically evaluate the success of the solution against the design specification

Aesthetics		
Design Specification	Method of Testing	Result
The font 'Roboto' must be used in the following sizes;  Headings: 25 - 30 Sub-headings: 20- 25 Paragraphs: 18 - 20	A user trial will be conducted and the processes used in the creation will simply be cross-checked.	All screen uses the font 'Roboto' with the mentioned font sizes. Hence, it passes.
The following color scheme must be used; 		The application uses all 5 colors, with an average score of 8/10 from user trials (Refer to Criterion D; Strand i). Hence, it passes.
Elements such as buttons and boxes must be curved at the edges by setting the border radius to either 50 or 75.		All elements are rounded at the edges with a border radius between 50 to 75. Hence, it passes.

There must be custom animations that make the application more dynamic.		There are 2 custom animations and according to user trails, it is believed to be dynamic by 100 % of the sample users (Refer to Criterion D; Strand i). Hence, it passes.	
Consumer			
Design Specification	Method of Testing	Result	
The application must cater to both the trainers and trainees of the club, by having different permissions.	A user trial will be conducted to see whether it caters to both types of users.	According to 100% of user in the trial, the application caters to both user types (Refer to Criterion D; Strand i). Hence, it passes.	
Ergonomics and Environment			
Design Specification	Method of Testing	Result	
The application must increase members scores during debates.	Analysis through graphical data collected after 1 week of using proposed solution.	 With the exception of 1 student, there has been an increase in the members debate scores. Hence, it passes. Note: Data was collected from the user trial form and was verified with club trainers to attain authentic data.	
The application must promote synergy and a sense of belonging.	User interview to interview whether synergy and a sense of belonging was promoted.	According to the user trial, 71.4 % of the sample users agreed that the application promoted a sense of belonging and synergy within the club. Hence, it passes.	
Safety and Size			
Design Specification	Method of Testing	Result	
User's must sign in using their school email id's	Performance Test	All users require an email id to sign in. Hence, it passes.	
The application must not share any user data with external parties of 3rd parties.	A risk assessment of the application will be conducted to test its safety.	Data is stored solely on google firebase and Airtable; Security practices To keep personal data safe, Firebase employs extensive security measures to minimize access: • Firebase restricts access to a select employees who have a business purpose to access personal data. • Firebase logs employee access to systems that contain personal data. • Firebase only permits access to personal data by employees who sign in with Google Sign-In and 2-factor authentication. Hence, it fails.	
Functionality			
Design Specification	Method of Testing	Result	
The Application must be divided into the trainers and the trainees' section, with a common sign in page.	A user trial will be conducted to see whether it caters to both types of users.	All users require an email id to sign in. Hence, it passes.	

The trainers must have the following screens; Home Debate settings	Performance testing, which is quantitative in nature will be conducted.	With reference to the user trials and performance test, it passes .
The trainees must have the following screens; Home Kudos		
The application must be available on both IOS and Android devices.		

Strand iii. explain how the solution could be improved

Weaknesses of the Product and Improvements

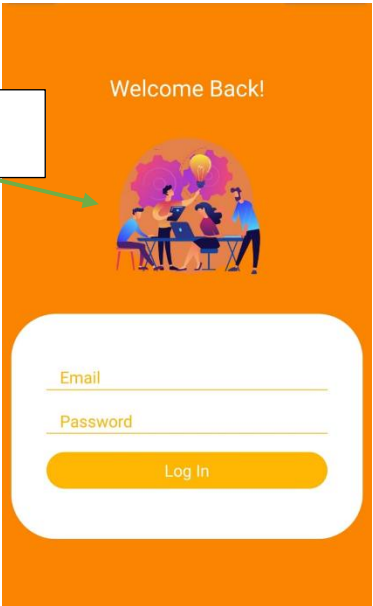
1. Response Time and retrieval of data: With reference to the User Trial, the response time for the elements of page when it started was higher than the expect response time. This caused the user to wait, for a longer period of time and can cause them to lose focus or become impatient due to our short attention spans. One of the causes is using JSON animations which is a UI factor.

Improvement	Benefits	Disadvantages
Store User Data Locally	This will decrease response time.	It may increase storage on the user’s phone, which may slow down other processes.
Remove JSON animations		Will make the application aesthetics less dynamic and static.

3. Custom Animations: The custom animations were not MUN specific and could be improved to be more relevant to the club. In addition, there were only two animations which did not make the application entirely dynamic.

Ex. Log In Screen

Generic teamwork animation



Improvement	Benefits	Disadvantages
Make MUN specific animations Ex. Students debating each other with the MUN logo	This will make it more relevant to the aesthetic purpose	It may increase response time as the file may be heavier.
Increase the number of custom animations.	Increase the dynamic power of the entire application	

4. Data Storage: As seen in the design specification to product comparison, the product failed in not sharing any user data with external parties of 3rd parties as the google firebase and Airtable privacy policy does not permit that.

Improvement	Benefits	Disadvantages
This can be improved by utilizing a different database platform with more strict privacy policies that comply with my design specification. Ex. Oracle	Safety of user Data	It may be costly.

5. Navigation: One issue with the application was that new users could not understand how to navigate around the application without the help of the developer. This may discourage MUN communities from using the application.

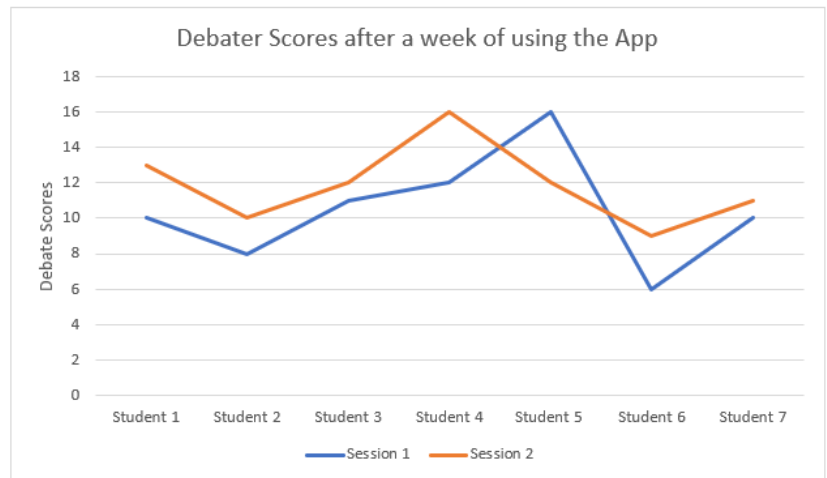
Improvement	Benefits	Disadvantages
Adding a screen with instructions of how to use and navigate around the application.	Ease of navigation.	It may increase file size.

Strand iv. explain the impact of the solution on the client/target audience.

User Trial

My product was able to address the lack of a sense of belonging within the MUN Club that my client, Amishi Srivastava felt due to a series of reasons (**Refer to Criterion A; Strand i**). According to the User Trial, 71.4 % of the club members agree that it promoted a sense of belonging and synergy within the club.

This allowed for a 33.3 % increase in debate score for my client from a score of 12 to a score of 16 by the end of the week of using the product. The same resulted in the all members of the club, with the exception of 1 member due to having a tough opponent.



Performance Testing

The performance test demonstrated a series of successes both quantitative and qualitative that satisfy the success criteria created in Criterion B. Therefore, meeting the client and target audience requirements.

Product Comparison

The proposed solution of a virtual application was highly relevant in the covid-19 context and in the current world where technology is the most efficient mean. By comparing the application to an existing product, 'HR Cloud', I was able to determine that they share many similarities with different purposes. Therefore, it is logical to assume that they would have similar benefits.

The benefits are demonstrated in testimonials from HR Cloud users. For example,

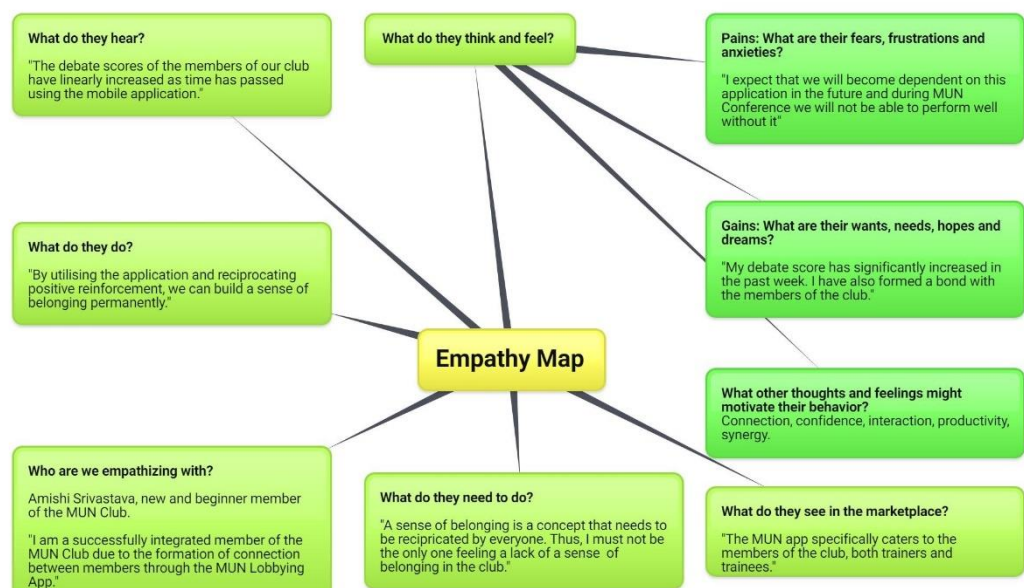
"HR Cloud's Onboard has helped our HR team save so much valuable time and effort while increasing data accuracy," by Kaylee Colins a HR analyst from Osmose.

Testimonial from sample user for MUN Lobbying App,

"The MUN Club Application allowed me to significantly increase my debate score through simply increasing interaction between club members that resulted in lobbying and eventually, promoted a sense of belonging."

Empathy Map

It is important to empathize with the client both before and after the product implementation. This can be demonstrated through the empathy map.



Global Context

The chosen exploration of the global context was addressed as the proposed solution helps develop a sense of belonging and amplifies identity formation within a Model UN environment in order to promote synergy.

Key and Related Concept

The development of social connections and bonds is one of the goals, alongside self-confidence and self-esteem as mentioned in Criterion A. With the involvement of the club members in debating and using the application, I was able to address the related concept of collaboration through mutual appreciation.

Statement of Inquiry

Ultimately, this allowed me to create the proposed solution with reference to the client's needs through the design specification that allowed me to remain objective and still empathize with them. Thus, promoting a sense of community and belonging with high debate scores.

ATL Skills

I was able to successfully address the ATL Skill of collaboration, as I had to interact with members of the club numerous times. In addition, excellent research skills were demonstrated by acquiring a variety of sources and types of information. Lastly, Self-management skills came into play to manage timelines efficiently and complete tasks.

OPVL Analysis

Analyzing my product with the origin, purpose, value, limitation technique; it was created for Model UN Student Clubs. It originates from the theory that appreciation from peers can demonstrate a sense of belonging and synergy in the form of higher debate scores. Thus, came the purpose with the concept of a mobile application that would virtually allow club members to lobby during debates. It has immense value as has demonstrated a linear increase in the debate scores of the club members after 1 week of use, which validates the virtuous cycle (Refer to Criterion A; strand i). Lastly, the two limitations that can be seen are in its functionality. Firstly, the response time of the application and secondly the need for club members to continuously use the application in order to see its benefits as it only temporarily instills a sense of community.

Conclusively, from the various testing methods I believe my product will have a positive impact on my client and target audience due to its success in various criteria.